AlphaFold provenance of the 720 parent–mutant pairs

Provenance supplemental catalogue

Record snapshot, 29 July 2026

What the table records

The data set contains 720 AlphaFold-derived parent structures. Each parent was retained and paired with a derivative in which every phenylalanine, tyrosine, tryptophan and histidine residue was replaced by alanine. The pair therefore contributes two structures: 720 parents and 720 aromatic-to-alanine derivatives, or 1,440 structures in total. No residue was inserted or deleted.

The stable AlphaFold DB model entity and the short protein description and source organism were resolved from the AlphaFold DB prediction API on 29 July 2026. The printed model entity has the form `AF-<UniProt>-F1`; unlike the downloadable file-version suffix, this identifier does not change when AlphaFold DB republishes an entry. The API sequence was checked against the parent residue axis for every row.

Residue counts and replacement sites come from an exact row-by-row comparison of `residues.npy` and `ala/residues.npy` in the `full720_20260716_2241` extraction. All 720 pairs have one chain, labelled A, with no insertion codes. A label such as `TYR-24` means that Tyr 24 in the parent was Ala 24 in the derivative. Across the full set, 2,469 aromatic sites were replaced; no other residue identities changed. The same residue counts and replacement lists were independently recovered from all 720 original AlphaFold PDBs and their generated mutant PDBs in the source archive.

The accompanying CSV retains the UniProt accession and entry name, taxonomy identifier, review status, AlphaFold DB model metadata, sequence checksums, lookup time, mutation count, and the two residue-axis paths used for each row. It also records SHA-256 hashes of the two source PDBs.

Table 1: AlphaFold parent structures and the aromatic residues replaced by alanine in their paired derivatives.

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A062V9G2-F1	Uncharacterized protein *Candidatus Methanoperedens nitroreducens*	48	TYR-4, HIS-10, PHE-33, HIS-40, TRP-43
AF-A0A075FL01-F1	peptide-methionine (R)-S-oxide reductase *uncultured marine thaumarchaeote AD1000_01_A07*	43	HIS-14, HIS-17, PHE-19, TYR-30
AF-A0A075FR84-F1	Nucleoside-diphosphate-sugar epimerases *uncultured marine thaumarchaeote AD1000_40_H03*	43	PHE-4, PHE-11, PHE-15, TYR-25, TYR-30
AF-A0A075FUC0-F1	Uncharacterized protein *uncultured marine thaumarchaeote AD1000_54_F06*	43	TYR-21
AF-A0A075FUI5-F1	Uncharacterized protein *uncultured marine thaumarchaeote AD1000_54_F06*	48	PHE-5, HIS-9, PHE-10, PHE-22, HIS-32, TYR-40, TYR-42, PHE-47
AF-A0A075G2N9-F1	peptide-methionine (R)-S-oxide reductase *uncultured marine thaumarchaeote KM3_02_H10*	50	TYR-6, HIS-21, HIS-24, PHE-26, TYR-37
AF-A0A075G604-F1	Anthranilate/para-aminobenzoate synthases component II *uncultured marine thaumarchaeote KM3_103_A05*	39	HIS-5, TYR-8, PHE-15, HIS-16, PHE-32
AF-A0A075G8S5-F1	phosphoribosyl-AMP cyclohydrolase *uncultured marine thaumarchaeote KM3_131_F04*	49	TYR-26, HIS-36, PHE-43, HIS-44
AF-A0A075G9H6-F1	Transcription factor, NikR *uncultured marine thaumarchaeote KM3_136_D12*	44	HIS-2, PHE-14, PHE-30
AF-A0A075GBG4-F1	Glutamine amidotransferase of anthranilate synthase (TrpG) *uncultured marine thaumarchaeote KM3_103_A05*	42	PHE-3, TYR-9, PHE-12, TYR-14
AF-A0A075GBP5-F1	Nucleotidyl transferase domain-containing protein *uncultured marine thaumarchaeote KM3_105_E03*	45	HIS-36, TRP-40
AF-A0A075GJJ0-F1	Uncharacterized protein *uncultured marine thaumarchaeote KM3_172_D05*	47	PHE-6, HIS-21, HIS-26, HIS-44
AF-A0A075GL14-F1	Uncharacterized protein *uncultured marine thaumarchaeote KM3_157_H08*	47	PHE-6, HIS-21, HIS-26, HIS-44
AF-A0A075GNQ9-F1	Transcription factor NikR nickel binding C-terminal domain-containing protein *uncultured marine thaumarchaeote KM3_183_D02*	44	HIS-2, PHE-14, PHE-30
AF-A0A075GPI2-F1	peptide-methionine (R)-S-oxide reductase *uncultured marine thaumarchaeote KM3_177_D01*	50	TYR-6, HIS-21, HIS-24, PHE-26, TYR-37
AF-A0A075GT79-F1	Transcription factor NikR nickel binding C-terminal domain-containing protein *uncultured marine thaumarchaeote KM3_183_D03*	44	HIS-2, PHE-14, PHE-30
AF-A0A075H126-F1	Uncharacterized protein *uncultured marine thaumarchaeote KM3_33_B12*	44	PHE-13
AF-A0A075H1U6-F1	peptide-methionine (R)-S-oxide reductase *uncultured marine thaumarchaeote KM3_42_G11*	43	HIS-14, HIS-17, PHE-19, TYR-30
AF-A0A075HED3-F1	peptide-methionine (R)-S-oxide reductase *uncultured marine thaumarchaeote KM3_57_D03*	50	TYR-6, HIS-21, HIS-24, PHE-26, TYR-37
AF-A0A075HHM8-F1	peptide-methionine (R)-S-oxide reductase *uncultured marine thaumarchaeote KM3_70_D07*	42	HIS-14, HIS-17, PHE-19, TYR-30
AF-A0A075HJ46-F1	peptide-methionine (R)-S-oxide reductase *uncultured marine thaumarchaeote KM3_74_C01*	50	TYR-6, HIS-21, HIS-24, PHE-26, TYR-37
AF-A0A075HVM8-F1	peptide-methionine (R)-S-oxide reductase *uncultured marine thaumarchaeote KM3_87_C09*	50	TYR-6, HIS-21, HIS-24, PHE-26, TYR-37, HIS-45
AF-A0A075I1Q2-F1	Rieske (2Fe-2S) domain-containing protein (HcaC, bphF) *uncultured marine thaumarchaeote SAT1000_09_C08*	42	HIS-6, PHE-7, TRP-8, HIS-9, TYR-10, TYR-20, TYR-29, TYR-39
AF-A0A075I3Q2-F1	Rieske (2Fe-2S) domain-containing protein (HcaC, bphF) *uncultured marine thaumarchaeote SAT1000_09_B08*	42	HIS-6, PHE-7, TRP-8, HIS-9, TYR-10, TYR-20, TYR-29, TYR-39
AF-A0A075I6E0-F1	Rieske (2Fe-2S) domain-containing protein (HcaC, bphF) *uncultured marine thaumarchaeote SAT1000_09_B07*	42	HIS-6, PHE-7, TRP-8, HIS-9, TYR-10, TYR-20, TYR-29, TYR-39
AF-A0A075I8Y7-F1	Rieske (2Fe-2S) domain-containing protein (HcaC, bphF) *uncultured marine thaumarchaeote SAT1000_09_C07*	42	HIS-6, PHE-7, TRP-8, HIS-9, TYR-10, TYR-20, TYR-29, TYR-39
AF-A0A075I9N7-F1	phosphoribosyl-AMP cyclohydrolase *uncultured marine thaumarchaeote SAT1000_10_G09*	48	TYR-26, HIS-36, PHE-43, HIS-44

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A075I9Q6-F1	tRNA-guanine transglycosylase (Tgt, QTRT1) *uncultured marine thaumarchaeote SAT1000_10_G09*	43	PHE-2, TYR-17, HIS-20, PHE-28, HIS-33
AF-A0A087RU44-F1	Nucleoside-diphosphate kinase protein *Marine Group I thaumarchaeote SCGC AAA799-P11*	24	PHE-7
AF-A0A087S4K0-F1	Uncharacterized protein *Marine Group I thaumarchaeote SCGC AAA799-B03*	42	PHE-19, PHE-24, TRP-26, PHE-33, TYR-40
AF-A0A0F2L3J9-F1	Glycerophosphodiester phosphodiesterase *Vulcanisaeta sp. AZ3*	44	PHE-5, HIS-7, TYR-13, PHE-22, HIS-38
AF-A0A0F8EH70-F1	Desulforedoxin *Methanosarcina sp. 2.H.A.1B.4*	41	TYR-10
AF-A0A0F8MQP1-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanosarcina mazei*	41	TYR-10
AF-A0A0U2L4M8-F1	Rubredoxin *Methanobrevibacter millerae*	36	PHE-3, TYR-10, HIS-12, PHE-21, PHE-33
AF-A0A0U3H9J4-F1	TsaA-like domain-containing protein *Thermococcus sp. 2319x1*	40	PHE-12, PHE-19, PHE-30
AF-A0A101DYV5-F1	NADH oxidase (NoxB-2) *Archaeoglobus fulgidus*	41	TYR-5, TYR-16
AF-A0A101DZD3-F1	Nitrogen regulatory protein P-II (GlnB-3) *Archaeoglobus fulgidus*	33	PHE-10, PHE-16
AF-A0A101E0G0-F1	Flavodoxin *Archaeoglobus fulgidus*	40	TYR-8, TRP-9, TRP-15, PHE-20, TYR-32, TYR-35
AF-A0A101GKX7-F1	Quinoprotein ethanol dehydrogenase *Methanoculleus marisnigri*	45	PHE-4, TYR-17, PHE-19, TYR-28, TYR-29, PHE-30, HIS-35, PHE-42
AF-A0A101GND4-F1	Uncharacterized protein family (UPF0150) *Methanoculleus marisnigri*	29	PHE-3, PHE-15, TRP-16, TYR-26
AF-A0A101IR95-F1	Cytidine kinase / inosine-guanosine kinase *Methanoculleus marisnigri*	50	HIS-7, HIS-12, PHE-14, HIS-23, TYR-27, HIS-31, TYR-34, PHE-35
AF-A0A101ISX8-F1	Glutamyl-tRNA(Gln) amidotransferase subunit E *Methanoculleus marisnigri*	32	TYR-3, HIS-15, PHE-25
AF-A0A117LQ45-F1	Aminotransferase *Methanoculleus marisnigri*	32	PHE-5, TYR-6, TRP-10, TYR-15
AF-A0A117MEV4-F1	Protein tyrosine phosphatase *Methanoculleus marisnigri*	38	PHE-7, HIS-11, TYR-22, TYR-27, TYR-31, PHE-34
AF-A0A117MF79-F1	UBA/THIF-type NAD/FAD binding protein *Methanoculleus marisnigri*	47	TYR-10, PHE-17, PHE-31, TYR-45
AF-A0A124F7Z6-F1	NADH oxidase (NoxB-2) *Archaeoglobus fulgidus*	41	TYR-5, TYR-16
AF-A0A126QX97-F1	Rubredoxin *methanogenic archaeon mixed culture ISO4-G1*	44	TYR-3, TYR-10, TYR-12, PHE-21, TRP-28, PHE-40
AF-A0A139CG29-F1	Homoisocitrate dehydrogenase *Methanohalophilus sp. T328-1*	37	PHE-26
AF-A0A1C8ZW37-F1	ATPase *Saccharolobus sp. A20*	50	TYR-4, TYR-27
AF-A0A1G6L254-F1	Regulatory protein, FmdB family *Natrinema hispanicum*	44	TYR-4, PHE-6, PHE-42
AF-A0A1G7QZ38-F1	Uncharacterized protein *Halorubrum xinjiangense*	49	HIS-8, PHE-11, PHE-31, PHE-38
AF-A0A1H1EH39-F1	Large ribosomal subunit protein eL39 *Halopelagius longus*	50	TRP-25, TRP-43
AF-A0A1I0QAW8-F1	Large ribosomal subunit protein eL39 *Natrinema salifodinae*	50	TRP-25, TRP-43
AF-A0A1L3Q2M2-F1	Desulfoferrodoxin *Methanohalophilus halophilus*	40	TYR-10
AF-A0A1L9C4R6-F1	Desulfoferrodoxin Dfx domain protein *Methanohalophilus portucalensis FDF-1*	40	TYR-10
AF-A0A1Q9NRY5-F1	Uncharacterized protein *Heimdallarchaeota archaeon (strain LC_2)*	50	TRP-9, HIS-21, TRP-22, PHE-31

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A1S6H833-F1	N-acetyltransferase domain-containing protein *Uncultured archaeon*	45	HIS-11, TYR-27, TYR-28, TYR-33, TYR-39
AF-A0A1S6H9M5-F1	Uncharacterized protein *Uncultured archaeon*	49	TYR-35
AF-A0A1V4TPG4-F1	Desulfoferrodoxin N-terminal domain-containing protein *Methanomassiliicoccales archaeon PtaB.Bin215*	39	TYR-8
AF-A0A1V4YX59-F1	Uncharacterized protein *Methanocella sp. PtaU1.Bin125*	37	TRP-16, TRP-32, PHE-33, HIS-36
AF-A0A1V4Z9N6-F1	Rubredoxin *Methanocella sp. PtaU1.Bin125*	35	TRP-3, TYR-10, HIS-12, PHE-31
AF-A0A219ANU8-F1	Uncharacterized protein *Methanobrevibacter sp. 87.7*	34	TRP-11, TYR-18
AF-A0A284VNA8-F1	Uncharacterized protein *Candidatus Methanoperedens nitroreducens*	40	PHE-10, TYR-16, HIS-32
AF-A0A285ESG6-F1	Polyphosphate kinase 2 (PPK2) *Methanohalophilus euhalobius*	41	TYR-9, TRP-12, TYR-22
AF-A0A285EYT8-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanohalophilus euhalobius*	40	TYR-10
AF-A0A2E0QIE5-F1	3-phosphoglycerate dehydrogenase *Nitrososphaerota archaeon*	39	PHE-6, HIS-11
AF-A0A2E4JTY0-F1	50S ribosomal protein L34 *Nitrososphaerota archaeon*	44	TYR-5, HIS-16, PHE-18
AF-A0A2G1WNH3-F1	superoxide dismutase *Halorubrum persicum*	48	TYR-3, TYR-10, TYR-12, HIS-18, TRP-26, HIS-27, HIS-28, HIS-31, HIS-32, TYR-35, TRP-39
AF-A0A2G9PKM8-F1	HicB family protein *Candidatus Woesearchaeota archaeon CG08_land_8_20_14_0_20_47_9*	45	PHE-11, TYR-14, TYR-42
AF-A0A2H9PSD1-F1	Stress response translation initiation inhibitor YciH *Candidatus Woesearchaeota archaeon CG_4_10_14_0_2_um_filter_33_10*	30	HIS-9, PHE-22
AF-A0A2P0QLX2-F1	5-formyltetrahydrofolate cyclo-ligase *Uncultured archaeon*	36	PHE-5, TYR-24, PHE-33
AF-A0A2R4ZQL9-F1	Succinate dehydrogenase/fumarate reductase iron-sulfur subunit *Desulfurococcaceae archaeon AG1*	46	HIS-14, HIS-25, TYR-39
AF-A0A2R6C3Z5-F1	Proteasome subunit beta *Candidatus Marsarchaeota G2 archaeon ECH_B_SAG-G06*	38	PHE-25, PHE-26, TYR-36
AF-A0A2R6C5B4-F1	PIN domain-containing protein *Candidatus Marsarchaeota G2 archaeon BE_D*	46	PHE-11, TYR-13, TYR-14, HIS-22, TRP-35, TYR-43
AF-A0A2R6C8T4-F1	CoA-binding domain-containing protein *Candidatus Marsarchaeota G2 archaeon BE_D*	42	PHE-23, HIS-24, TYR-31
AF-A0A2R6CA67-F1	Aspartate/ornithine carbamoyltransferase carbamoyl-P binding domain-containing protein *Candidatus Marsarchaeota G2 archaeon BE_D*	45	HIS-7, PHE-28, HIS-30, PHE-38
AF-A0A2R6CAA9-F1	Metal-dependent hydrolase *Candidatus Marsarchaeota G2 archaeon BE_D*	32	TYR-6, HIS-9, PHE-12, PHE-26
AF-A0A2R6D1W6-F1	Large ribosomal subunit protein eL39 *Halobacteriales archaeon QH_2_66_30*	50	TRP-25, TRP-43
AF-A0A2R6EXE0-F1	Plastocyanin *Halobacteriales archaeon QH_7_66_37*	35	PHE-3, HIS-5, PHE-7, TYR-13, TYR-15, PHE-16, HIS-20
AF-A0A2R6LXP5-F1	Phosphoribosylamine–glycine ligase *Halobacteriales archaeon SW_12_67_38*	46	HIS-15, PHE-29, PHE-46
AF-A0A2R6MFM1-F1	Aspartate-semialdehyde dehydrogenase *Halobacteriales archaeon SW_12_67_38*	45	TYR-17, HIS-22, TRP-44
AF-A0A2R6MFZ7-F1	DNA-binding protein *Halobacteriales archaeon SW_12_67_38*	48	PHE-3, PHE-12, PHE-16, TYR-48
AF-A0A2V3I199-F1	Cold-shock protein *Methanobacteriota archaeon*	27	PHE-9

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A2V3JFQ8-F1	Rubredoxin *Methanophagales archaeon*	45	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A2V3JMT2-F1	Ryanodine receptor Ryr *ANME-2 cluster archaeon*	45	HIS-8, TRP-12, TYR-22, PHE-33, PHE-39, TYR-42
AF-A0A2V4GT18-F1	Uncharacterized protein *Thermoplasma sp. Kam2015*	42	PHE-37, PHE-40, TRP-41
AF-A0A2V4H8E0-F1	UPF0033 domain-containing protein *Thermoplasma sp. Kam2015*	44	TYR-5, TRP-18, PHE-30, TYR-35, TYR-36
AF-A0A345DZ50-F1	Large ribosomal subunit protein eL39 *Haloplanus rubicundus*	50	TRP-25, TRP-43
AF-A0A367I6A3-F1	phosphoglycerate mutase (2,3-diphosphoglycerate-dependent) *Candidatus Poseidoniales archaeon*	42	HIS-9, TRP-14, PHE-20, TRP-23
AF-A0A371MR54-F1	Ferredoxin *Haloferax sp. Atlit-6N*	46	TYR-6, TYR-9, TRP-17, PHE-25, TYR-38
AF-A0A371MRP4-F1	Cysteine methyltransferase *Haloferax sp. Atlit-6N*	39	PHE-12, HIS-16
AF-A0A371MS07-F1	Diaminopimelate decarboxylase *Haloferax sp. Atlit-6N*	43	TYR-2, TYR-4, TYR-10
AF-A0A371MSL8-F1	Cell division protein *Haloferax sp. Atlit-6N*	44	TYR-22, PHE-30
AF-A0A371MTC3-F1	Chromosome segregation protein *Haloferax sp. Atlit-6N*	40	HIS-20
AF-A0A3A5VK63-F1	Carboxypeptidase M32 *Candidatus Poseidoniales archaeon*	46	TYR-4, PHE-7, HIS-15, HIS-23, TRP-26, TRP-46
AF-A0A3A5VSI2-F1	ATP:cob(I)alamin adenosyltransferase *Candidatus Poseidoniales archaeon*	48	HIS-8, PHE-32
AF-A0A3A5W4U3-F1	Riboflavin synthase *Candidatus Poseidoniales archaeon*	44	HIS-1, PHE-12
AF-A0A3A5WVH7-F1	5-(Carboxyamino)imidazole ribonucleotide mutase *Candidatus Poseidoniales archaeon*	48	PHE-26, TYR-30, HIS-38, PHE-43
AF-A0A3M0CXD9-F1	Large ribosomal subunit protein eL39 *Haloplanus aerogenes*	50	TRP-25, TRP-43
AF-A0A3M1LFP0-F1	PIN domain nuclease *Nitrososphaerota archaeon*	42	PHE-9, PHE-13, PHE-25, PHE-41
AF-A0A3M1LH11-F1	YjbQ family protein *Nitrososphaerota archaeon*	34	TYR-7, PHE-10
AF-A0A3M1LHW0-F1	Type II methionyl aminopeptidase *Nitrososphaerota archaeon*	33	TYR-3, TYR-13, HIS-19
AF-A0A3M1LK14-F1	Blue (type 1) copper domain-containing protein *Nitrososphaerota archaeon*	32	PHE-3, TYR-11, TYR-12, TYR-13, HIS-20, PHE-29
AF-A0A3M1LK61-F1	Nitroreductase *Nitrososphaerota archaeon*	47	PHE-3, PHE-4, HIS-11, TYR-16, TYR-45
AF-A0A3M1LLL7-F1	Thymidylate kinase *Nitrososphaerota archaeon*	50	TYR-27, PHE-32, PHE-34, TYR-37, PHE-48
AF-A0A3M1LN27-F1	Acylphosphatase *Nitrososphaerota archaeon*	37	HIS-7, TYR-18, PHE-19, HIS-30, TRP-35
AF-A0A3M1LPS3-F1	Zinc-ribbon domain-containing protein *Nitrososphaerota archaeon*	23	TYR-7, PHE-15
AF-A0A3M9LMY0-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanohalophilus sp. RSK*	40	TYR-10
AF-A0A3M9ZV07-F1	CTP synthase (glutamine hydrolyzing) *Nitrosopumilus sp. D6*	49	PHE-1, HIS-19, PHE-21, TYR-22, PHE-27, HIS-28, PHE-31, PHE-37, PHE-42, PHE-45
AF-A0A3M9ZZ00-F1	CoA-binding protein *Nitrosopumilus sp. H13*	41	HIS-29, TYR-30, TYR-34, HIS-38, TYR-40
AF-A0A3R7DRJ2-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Candidatus Bathyarchaeota archaeon*	38	TYR-8
AF-A0A3R7VPW5-F1	4Fe-4S Mo/W bis-MGD-type domain-containing protein *Methanocalculus sp. MSAO_Arc1*	40	TYR-5, TYR-12, TYR-34, HIS-35

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A421DU79-F1	NADPH:quinone oxidoreductase *Haloarcula sp. Atlit-120R*	49	PHE-9, PHE-15, HIS-18, HIS-35, TYR-44
AF-A0A432IOP7-F1	Amidohydrolase 3 domain-containing protein *Candidatus Poseidoniales archaeon*	43	HIS-15, PHE-24, TYR-34, HIS-41
AF-A0A432I6S4-F1	Serine hydroxymethyltransferase-like domain-containing protein *Candidatus Poseidoniales archaeon*	40	TYR-34
AF-A0A495R914-F1	ABC transmembrane type-1 domain-containing protein *Haloarcula quadrata*	39	HIS-36
AF-A0A495R9V0-F1	Transporter *Haloarcula quadrata*	41	PHE-15, PHE-23, TRP-38
AF-A0A497IHI4-F1	tRNA (Pseudouridine(54)-N(1))-methyltransferase TrmY *Methanosarcinales archaeon*	28	PHE-4, PHE-16
AF-A0A497KMA1-F1	Desulfoferrodoxin *Candidatus Bathyarchaeota archaeon*	38	TYR-8
AF-A0A497PY88-F1	Desulfoferrodoxin *Thorarchaeota archaeon (strain OWC)*	40	TYR-8, TYR-37
AF-A0A4U7F0R4-F1	superoxide dismutase *Halorubrum sp. SD626R*	50	TYR-3, TYR-10, TYR-12, HIS-18, TRP-26, HIS-27, HIS-28, HIS-31, HIS-32, TYR-35, TRP-39
AF-A0A4U7F4A3-F1	Superoxide dismutase *Halorubrum sp. SD626R*	40	HIS-2, TYR-4, TYR-5, HIS-6, TYR-8, PHE-15, PHE-19, PHE-20, TRP-25, TYR-32, PHE-39
AF-A0A510DZD2-F1	Uncharacterized protein *Sulfuracidifex tepidarius*	46	TRP-6, HIS-31, HIS-36
AF-A0A519BR66-F1	Proteasome assembly chaperone family protein *uncultured DHVE6 group euryarchaeote*	47	PHE-18, PHE-23, PHE-31, HIS-35, TYR-43, TYR-45
AF-A0A519BS09-F1	4a-hydroxytetrahydrobiopterin dehydratase *uncultured DHVE6 group euryarchaeote*	48	HIS-11, HIS-12, TRP-19, TYR-21, TYR-27, HIS-29, PHE-39
AF-A0A519BYY9-F1	Lysine biosynthesis enzyme LysX *Nitrososphaerota archaeon*	46	HIS-14, PHE-22, TYR-40
AF-A0A519C2G9-F1	peptide-methionine (S)-S-oxide reductase *Nitrososphaerota archaeon*	32	PHE-5, PHE-10, TRP-11, HIS-12, PHE-17, TYR-30
AF-A0A519CKS0-F1	1-(5-phosphoribosyl)-5-[(5-phosphoribosylamino)methylideneamino]imidazole-4-carboxamide isomerase *Nitrososphaerota archaeon*	39	TYR-18, TYR-28, TRP-39
AF-A0A519CLK2-F1	TIGR00269 family protein *Nitrososphaerota archaeon*	47	TYR-12, TYR-16, PHE-26, TYR-41, TYR-46
AF-A0A519CU77-F1	C-methyltransferase domain-containing protein *Nitrososphaerota archaeon*	40	TRP-10, TYR-12, PHE-23, PHE-30
AF-A0A520K3A4-F1	Gfo/Idh/MocA family oxidoreductase *Methanosarcinales archaeon*	50	HIS-15, TYR-19, TYR-29, PHE-31, TYR-44
AF-A0A520K6T6-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanosarcinales archaeon*	37	TYR-8
AF-A0A520KLZ7-F1	pyruvate, water dikinase *Methanonatronarchaeia archaeon*	48	TRP-5, PHE-38, PHE-45, PHE-48
AF-A0A523ASN7-F1	Gamma carbonic anhydrase family protein *Hadesarchaea archaeon*	45	TYR-4, PHE-6, HIS-14, TRP-38
AF-A0A523AT74-F1	Histidinol dehydrogenase *Hadesarchaea archaeon*	43	PHE-1, TYR-32, HIS-34
AF-A0A523AVT4-F1	Formylmethanofuran–tetrahydromethanopterin N-formyltransferase *Hadesarchaea archaeon*	47	PHE-12, PHE-16, TRP-32, PHE-43
AF-A0A523AYL0-F1	23S rRNA (Uridine(2552)-2'-O)-methyltransferase *Hadesarchaea archaeon*	46	TYR-3, PHE-9, TYR-18, TRP-37
AF-A0A523AYZ6-F1	CooT family nickel-binding protein *Hadesarchaea archaeon*	30	TYR-1, HIS-25
AF-A0A523BEK8-F1	DNA-binding protein *Candidatus Methanomethylicota archaeon*	36	TRP-6, TYR-31

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A523SSL6-F1	NDP-sugar synthase *Candidatus Bathyarchaeota archaeon*	34	HIS-10, TYR-19
AF-A0A524FGV4-F1	Rubredoxin *Promethearchaeota archaeon*	44	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A533U6D3-F1	NAD(P)-dependent oxidoreductase *Nitrososphaerota archaeon*	49	PHE-11
AF-A0A533UJM1-F1	Cysteinyl-tRNA synthetase *Nitrososphaerota archaeon*	41	PHE-5, PHE-8, PHE-15, PHE-18, HIS-22, PHE-26, TYR-33, TYR-35, HIS-37, HIS-40
AF-A0A533UWQ2-F1	3-phosphoglycerate dehydrogenase *Nitrososphaerota archaeon*	46	HIS-18, PHE-43
AF-A0A533V157-F1	Universal stress protein *Nitrososphaerota archaeon*	36	PHE-15, TYR-20, PHE-24, TYR-28
AF-A0A533V2Q5-F1	Fructose 1,6-bisphosphatase *Nitrososphaerota archaeon*	43	HIS-18, TYR-43
AF-A0A533V4X6-F1	Acetyl-CoA carboxylase biotin carboxylase subunit *Nitrososphaerota archaeon*	50	TYR-35, HIS-44
AF-A0A533V523-F1	Sulfurtransferase *Nitrososphaerota archaeon*	48	TYR-5, HIS-14, TRP-16, PHE-17, TYR-21, TYR-25, TYR-31, TRP-35, TRP-38
AF-A0A533VIP4-F1	Methionine synthase *Nitrososphaerota archaeon*	35	PHE-16
AF-A0A533VJR1-F1	DNA-3-methyladenine glycosylase *Nitrososphaerota archaeon*	37	HIS-9, TYR-21, HIS-30, TYR-32
AF-A0A533VSR0-F1	NAD-dependent malic enzyme *Nitrososphaerota archaeon*	45	HIS-12, TYR-37
AF-A0A533VW71-F1	Blue (type 1) copper domain-containing protein *Nitrososphaerota archaeon*	41	TYR-11, PHE-13, PHE-15, TYR-21, TYR-23, TYR-27, HIS-28, TRP-30, HIS-32
AF-A0A533W8U5-F1	PIG-L family deacetylase *Nitrososphaerota archaeon*	27	HIS-9
AF-A0A533WKQ8-F1	SDR family NAD(P)-dependent oxidoreductase *Nitrososphaerota archaeon*	50	PHE-18, PHE-24
AF-A0A533WM15-F1	Translation factor Sua5 *Nitrososphaerota archaeon*	49	PHE-6, PHE-28, TYR-34, TYR-41
AF-A0A533WMV9-F1	Aminodeoxychorismate/anthranilate synthase component II *Nitrososphaerota archaeon*	41	HIS-8, TYR-11, PHE-18, HIS-19, PHE-35
AF-A0A533WY04-F1	Dihydrolipoamide dehydrogenase *Nitrososphaerota archaeon*	48	HIS-5, HIS-11, HIS-24, HIS-31, HIS-33, PHE-47
AF-A0A533WZX1-F1	Acyl-CoA thioesterase *Nitrososphaerota archaeon*	50	PHE-17, HIS-24, HIS-36
AF-A0A533X1P5-F1	O-methyltransferase *Nitrososphaerota archaeon*	49	PHE-10, HIS-13, TYR-14, TYR-21, HIS-24
AF-A0A533XGD3-F1	Uracil phosphoribosyltransferase *Nitrososphaerota archaeon*	46	TYR-5, TYR-10, PHE-15, TYR-16, TYR-29, PHE-42, TYR-46
AF-A0A533XLT4-F1	Ester cyclase *Nitrososphaerota archaeon*	32	PHE-2, TRP-8, TYR-20
AF-A0A538P2N2-F1	Phosphoribosylamine-glycine ligase *Nitrososphaerota archaeon*	43	PHE-3, PHE-5, TRP-17, HIS-24, PHE-28, TYR-29, PHE-42
AF-A0A538P8P9-F1	Biotin carboxylation domain-containing protein *Nitrososphaerota archaeon*	43	PHE-2, PHE-32, HIS-41
AF-A0A538P9J5-F1	M20/M25/M40 family metallo-hydrolase *Nitrososphaerota archaeon*	48	PHE-11, HIS-20, TRP-25
AF-A0A550D5K8-F1	NAD(P)-dependent oxidoreductase *Sulfolobus sp. F3*	49	PHE-26
AF-A0A550GLI6-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Candidatus Bathyarchaeota archaeon*	38	TYR-8
AF-A0A558GAE9-F1	Large ribosomal subunit protein eL39 *Haloferax volcanii*	50	TRP-25, TRP-43
AF-A0A5C9ESF5-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Promethearchaeota archaeon*	39	TYR-8, PHE-24

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A5E4HI60-F1	PIN domain-containing protein *Uncultured archaeon*	44	PHE-4, PHE-10, TYR-13, HIS-21, HIS-26, TYR-36, PHE-40
AF-A0A5E4HJN0-F1	Tetratricopeptide repeat protein *Uncultured archaeon*	39	PHE-18
AF-A0A5E4HP61-F1	PD-(D/E)XK nuclease superfamily protein *Uncultured archaeon*	50	TYR-3, PHE-14, TYR-15, TYR-18, TYR-23, PHE-25, TYR-30, PHE-34, PHE-43
AF-A0A5E4I7H3-F1	Bacterial transferase hexapeptide (Six repeats) *Uncultured archaeon*	37	HIS-5, TYR-9, PHE-29
AF-A0A5E4IG34-F1	CobW/HypB/UreG nucleotide-binding domain-containing protein *Uncultured archaeon*	38	HIS-20
AF-A0A5E4IH91-F1	PIN domain-containing protein *Uncultured archaeon*	46	TYR-2, TRP-9, PHE-25, PHE-39, TYR-42
AF-A0A5E4IJB0-F1	DNA mismatch repair protein MutL *Uncultured archaeon*	44	TYR-14, HIS-20, HIS-24, PHE-41
AF-A0A5E4ILE0-F1	acylphosphatase *Uncultured archaeon*	48	TYR-7, TYR-17, PHE-28
AF-A0A5E4ISJ4-F1	PAC2 family protein *Uncultured archaeon*	39	HIS-7, HIS-28, HIS-36
AF-A0A5E4J1K7-F1	Uncharacterized protein *Uncultured archaeon*	37	PHE-7, TRP-12, PHE-24, TYR-29, TYR-30, TYR-33, HIS-37
AF-A0A5E4J7J9-F1	UvrABC system protein A *Uncultured archaeon*	47	HIS-12, PHE-26, PHE-42, TYR-46
AF-A0A5E4JHA9-F1	Uncharacterized protein *Uncultured archaeon*	49	TYR-39
AF-A0A5E4K421-F1	Uncharacterized protein *Uncultured archaeon*	32	PHE-29
AF-A0A5E4K5U7-F1	Putative NH(3)-dependent NAD(+) synthetase *Uncultured archaeon*	46	TYR-5, PHE-15, HIS-17, TYR-43
AF-A0A5E4K9E0-F1	DUF86 domain-containing protein *Uncultured archaeon*	41	HIS-7, TYR-9, PHE-10, TRP-18
AF-A0A5E4KHE1-F1	Heterodisulfide reductase subunit A-like protein *Uncultured archaeon*	42	TYR-6, HIS-9, HIS-26, PHE-41
AF-A0A5E4KT12-F1	Uncharacterized protein *Uncultured archaeon*	32	PHE-5, PHE-21, PHE-22
AF-A0A5E4L400-F1	Desulfoferrodoxin N-terminal domain-containing protein *Uncultured archaeon*	40	TYR-10
AF-A0A5E4LE60-F1	Uncharacterized protein *Uncultured archaeon*	44	PHE-4, TYR-15, TYR-37
AF-A0A5N5UDU8-F1	Large ribosomal subunit protein eL39 *Halosegnis rubeus*	50	TRP-25, TRP-43
AF-A0A660HR30-F1	Type II toxin-antitoxin system HicB family antitoxin *Methanosarcina flavescens*	45	TYR-5, PHE-7, HIS-12, TRP-17, TYR-18, TRP-21
AF-A0A662PMX3-F1	Adenosine monophosphate-protein transferase *Nitrososphaerota archaeon*	45	TYR-20, PHE-37, TYR-43, PHE-45
AF-A0A662PNS1-F1	Ribonuclease H *Nitrososphaerota archaeon*	50	PHE-7, TYR-23, PHE-39, TYR-50
AF-A0A662PVT5-F1	Phosphohistidine phosphatase SixA *Nitrososphaerota archaeon*	41	TYR-4, HIS-8, PHE-36
AF-A0A662PX72-F1	2-deoxyribose-5-phosphate aldolase *Nitrososphaerota archaeon*	47	TRP-4, HIS-13, HIS-19, PHE-35, PHE-37, TYR-46
AF-A0A662Q049-F1	UDP-glucose/GDP-mannose dehydrogenase N-terminal domain-containing protein *Nitrososphaerota archaeon*	43	TYR-11, PHE-16, PHE-25, TYR-28
AF-A0A662Q1I9-F1	Homoserine kinase *Nitrososphaerota archaeon*	41	PHE-16, PHE-19, TYR-27
AF-A0A662Q1V9-F1	ATP-binding protein *Nitrososphaerota archaeon*	31	PHE-7, TYR-15, TYR-18, HIS-25, TYR-29
AF-A0A662Q584-F1	Alcohol dehydrogenase *Nitrososphaerota archaeon*	43	TYR-5, PHE-6, TYR-17, TYR-33

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A662Q601-F1	CopG family transcriptional regulator *Nitrososphaerota archaeon*	39	TYR-18, TRP-38
AF-A0A662QA52-F1	Cof-type HAD-IIB family hydrolase *Nitrososphaerota archaeon*	46	HIS-24, PHE-30, TRP-37
AF-A0A662QCF6-F1	Hydrolase *Nitrososphaerota archaeon*	41	PHE-10, HIS-17
AF-A0A662QCK9-F1	Acetate kinase *Nitrososphaerota archaeon*	45	TYR-15, TYR-40, HIS-43
AF-A0A662QEP8-F1	Diphthine synthase *Nitrososphaerota archaeon*	41	PHE-6, TYR-21, PHE-37, TRP-39
AF-A0A662QXS3-F1	Desulfoferrodoxin N-terminal domain-containing protein *Methanosarcinales archaeon*	38	TYR-8
AF-A0A662R080-F1	3-isopropylmalate dehydratase *Methanosarcinales archaeon*	25	TRP-8, PHE-10, TYR-24
AF-A0A662R097-F1	Precorrin-4 C(11)-methyltransferase *Methanosarcinales archaeon*	35	TYR-6, PHE-7, TYR-34
AF-A0A662R0I2-F1	Winged helix-turn-helix transcription repressor HrcA DNA-binding domain-containing protein *Methanosarcinales archaeon*	41	TYR-18
AF-A0A662R243-F1	Mannosyl-3-phosphoglycerate phosphatase *Methanosarcinales archaeon*	44	TRP-3, PHE-6, TYR-18, PHE-20, TRP-29, HIS-34, PHE-39
AF-A0A662R2Z3-F1	ATP phosphoribosyltransferase *Methanosarcinales archaeon*	38	TYR-1, TYR-2, HIS-38
AF-A0A662R4C9-F1	Imidazole glycerol phosphate synthase subunit HisF *Methanosarcinales archaeon*	47	HIS-18, PHE-27, TYR-32, TRP-37, TYR-43
AF-A0A662R555-F1	DegT/DnrJ/EryC1/StrS aminotransferase family protein *Methanosarcinales archaeon*	43	TYR-16, TRP-25, TYR-32, PHE-36, PHE-40, TYR-43
AF-A0A662RAM6-F1	Type II toxin-antitoxin system HicB family antitoxin *Methanosarcinales archaeon*	48	HIS-4, PHE-5, TYR-9, TRP-10, TRP-15, TYR-16, PHE-27, TYR-44
AF-A0A662RK02-F1	Desulfoferrodoxin *Methanosarcinales archaeon*	37	TYR-8
AF-A0A662RTJ4-F1	Acetolactate synthase small subunit *Methanosarcinales archaeon*	37	PHE-12, HIS-18, PHE-19, PHE-25
AF-A0A662RX65-F1	Molybdenum cofactor biosynthesis protein *Methanosarcinales archaeon*	34	PHE-4, HIS-27, HIS-31
AF-A0A662T3H2-F1	NADH:flavin oxidoreductase *Candidatus Geothermarchaeota archaeon*	44	TYR-13, HIS-42, TYR-43
AF-A0A662T6Q3-F1	acylphosphatase *Candidatus Geothermarchaeota archaeon*	38	TYR-4, TYR-6, PHE-19, PHE-22, PHE-28, TYR-35
AF-A0A662T947-F1	Uncharacterized protein *Candidatus Geothermarchaeota archaeon*	46	PHE-10, TRP-32, PHE-37, TRP-41, TYR-44, PHE-46
AF-A0A662UDV1-F1	Chorismate-binding protein *Thermoprotei archaeon*	46	TRP-18, PHE-33, HIS-39, TYR-40, PHE-41
AF-A0A6A7LI12-F1	Multicopper oxidase domain-containing protein *Nitrososphaeraceae archaeon*	43	PHE-14, PHE-17, PHE-21, TYR-23, HIS-24, HIS-26, PHE-29, HIS-30
AF-A0A6A7LI67-F1	SDR family oxidoreductase *Nitrososphaeraceae archaeon*	39	TRP-4, HIS-8, PHE-16, PHE-24, PHE-31
AF-A0A6A7LJ60-F1	Pyridoxamine 5'-phosphate oxidase N-terminal domain-containing protein *Nitrososphaeraceae archaeon*	40	PHE-9, PHE-15, HIS-29, TRP-34, PHE-35
AF-A0A6A7LLL2-F1	NADH-quinone oxidoreductase subunit NuoK *Nitrososphaeraceae archaeon*	41	TYR-7, PHE-15, TYR-20, PHE-33
AF-A0A6A7LMN5-F1	ASCH domain-containing protein *Nitrososphaeraceae archaeon*	37	TYR-10, TRP-26, PHE-30, PHE-34
AF-A0A6A8AGZ0-F1	FAD-binding protein *Methanosarcinales archaeon*	33	TYR-6
AF-A0A6A8AI96-F1	Rubredoxin *Methanosarcinales archaeon*	45	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A6A8AP17-F1	NAD-dependent epimerase/dehydratase family protein *Methanosarcinales archaeon*	39	PHE-11, HIS-15, TYR-25, PHE-33, TYR-36, TYR-37

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A6A9T401-F1	KEOPS complex Pcc1-like subunit *Halapricum sp. CBA1109*	46	HIS-5, HIS-12
AF-A0A6B0HK18-F1	Large ribosomal subunit protein eL39 *Natrialba sp. INN-245*	50	TRP-25, TRP-43
AF-A0A6B9QZA6-F1	coenzyme-B sulfoethylthiotransferase *uncultured Candidatus Methanoperedens sp*	50	TYR-4, HIS-9, TYR-23, HIS-38, PHE-39, TRP-49
AF-A0A6G3LQ52-F1	Uncharacterized protein *Acidilobus sp*	46	TYR-2, HIS-5
AF-A0A6V8C9N9-F1	Thioredoxin *Candidatus Poseidoniales archaeon*	50	PHE-2, TRP-3, TRP-6, HIS-39
AF-A0A6V8CBP4-F1	Endonuclease III *Candidatus Poseidoniales archaeon*	47	PHE-2, TYR-25, HIS-46, HIS-47
AF-A0A6V8CC87-F1	SDR family oxidoreductase *Candidatus Poseidoniales archaeon*	40	TYR-19, TYR-25, TRP-37
AF-A0A6V8CDD8-F1	NAD(P)(+) transhydrogenase (Re/Si-specific) subunit alpha *Candidatus Poseidoniales archaeon*	50	PHE-30, HIS-44, TYR-49
AF-A0A6V8CLY9-F1	NADPH:quinone reductase *Candidatus Poseidoniales archaeon*	43	TRP-5, TYR-6, HIS-17
AF-A0A6V8CMT5-F1	Phosphoribosylamine–glycine ligase *Candidatus Poseidoniales archaeon*	49	HIS-15, HIS-31, HIS-41, TYR-42, HIS-46
AF-A0A6V8CMX5-F1	Malate dehydrogenase *Candidatus Poseidoniales archaeon*	47	HIS-2, TYR-12
AF-A0A6V8CN64-F1	nucleoside-diphosphate kinase *Candidatus Poseidoniales archaeon*	33	TYR-5, PHE-25
AF-A0A6V8CNG8-F1	Adenosylhomocysteinase *Candidatus Poseidoniales archaeon*	48	HIS-7, TRP-28, TRP-43
AF-A0A6V8CNU4-F1	Glutamate dehydrogenase *Candidatus Poseidoniales archaeon*	45	HIS-7, TYR-11, TYR-17, TYR-23, PHE-25, PHE-32
AF-A0A6V8CPX8-F1	peptide-methionine (R)-S-oxide reductase *Candidatus Poseidoniales archaeon*	46	HIS-1, HIS-16, HIS-19, PHE-21, HIS-28, TYR-33, PHE-42
AF-A0A6V8CQS9-F1	Malate dehydrogenase *Candidatus Poseidoniales archaeon*	48	PHE-4, HIS-20, HIS-24
AF-A0A6V8CR43-F1	Uncharacterized protein *Candidatus Poseidoniales archaeon*	36	TYR-2, TYR-3, TYR-8, HIS-9, PHE-12, PHE-23
AF-A0A6V8CTA7-F1	FAD synthase *Candidatus Poseidoniales archaeon*	43	PHE-10, HIS-14, HIS-17, HIS-19, TYR-20, HIS-37
AF-A0A6V8CUM7-F1	Trk system potassium transporter TrkA *Candidatus Poseidoniales archaeon*	49	PHE-4, TYR-22
AF-A0A6V8CZK2-F1	FAD-binding protein *Candidatus Poseidoniales archaeon*	34	TYR-18
AF-A0A6V8D2T6-F1	Class II fumarate hydratase *Candidatus Poseidoniales archaeon*	49	HIS-28, PHE-40, TYR-45
AF-A0A6V8D831-F1	N(6)-L-threonylcarbamoyladenine synthase *Candidatus Poseidoniales archaeon*	50	TYR-11, HIS-44, HIS-48
AF-A0A6V8DL76-F1	UDP-N-acetylglucosamine pyrophosphorylase *Candidatus Poseidoniales archaeon*	41	TRP-25, HIS-34, HIS-40
AF-A0A6V8DQK0-F1	leucine–tRNA ligase *Candidatus Poseidoniales archaeon*	38	PHE-2, TYR-4, HIS-10, HIS-13, TYR-17, PHE-26, TYR-36
AF-A0A6V8DTT4-F1	nucleoside-diphosphate kinase *Candidatus Poseidoniales archaeon*	34	TYR-5, PHE-25
AF-A0A6V8DZ60-F1	EVE domain-containing protein *Candidatus Poseidoniales archaeon*	40	TYR-4, TRP-5, TYR-14, TYR-16, HIS-26, TRP-27, TYR-33
AF-A0A6V8E1N8-F1	Ribonuclease Z *Candidatus Poseidoniales archaeon*	45	HIS-5, PHE-25, PHE-40, TYR-44
AF-A0A6V8E4D6-F1	Lipoyl-binding domain-containing protein *Candidatus Poseidoniales archaeon*	49	HIS-18, TYR-33
AF-A0A6V8E5D1-F1	Dihydroneopterin aldolase *Candidatus Poseidoniales archaeon*	38	PHE-2, TYR-9, HIS-15, TYR-18, PHE-20, HIS-22, PHE-28, TRP-33, HIS-38
AF-A0A6V8E729-F1	Glucose-1-phosphate thymidylyltransferase *Candidatus Poseidoniales archaeon*	47	PHE-2, TRP-31, TYR-40, HIS-46

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A6V8ERQ8-F1	Adenylosuccinate synthase *Candidatus Poseidoniales archaeon*	42	PHE-2, HIS-3, HIS-10, TYR-19, HIS-22, TYR-30
AF-A0A6V8ESH0-F1	Thioredoxin-disulfide reductase *Candidatus Poseidoniales archaeon*	41	TYR-3, TYR-8, PHE-19, TYR-22, PHE-37, TYR-40
AF-A0A6V8ESU9-F1	7-carboxy-7-deazaguanine synthase QueE *Candidatus Poseidoniales archaeon*	42	TYR-5, PHE-11, HIS-12, PHE-19, HIS-20, HIS-25, PHE-27, PHE-30, TRP-39
AF-A0A6V8EWF7-F1	DNA-formamidopyrimidine glycosylase *Candidatus Poseidoniales archaeon*	47	PHE-35, PHE-37, PHE-41
AF-A0A6V8EWP6-F1	HEAT repeat domain-containing protein *Candidatus Poseidoniales archaeon*	50	HIS-6, TYR-10, HIS-34, HIS-39
AF-A0A6V8EXG4-F1	peptidylprolyl isomerase *Candidatus Poseidoniales archaeon*	50	HIS-2, PHE-17, HIS-18, PHE-28, PHE-36, TYR-37, HIS-41, PHE-42, HIS-43, PHE-49
AF-A0A7C0XQX0-F1	Glycine cleavage system protein H *Acidilobales archaeon*	48	TYR-13, TYR-21, TRP-27, TYR-42
AF-A0A7C0XZN5-F1	2Fe-2S iron-sulfur cluster binding domain-containing protein *Acidilobales archaeon*	48	PHE-5
AF-A0A7C0Y320-F1	Mevalonate kinase *Acidilobales archaeon*	36	HIS-2, PHE-14, HIS-17, TYR-21, TYR-35
AF-A0A7C1BUZ8-F1	ATP-binding protein *Acidilobales archaeon*	31	TYR-7, HIS-14, TYR-15
AF-A0A7C1M065-F1	N-acetyl-gamma-glutamyl-phosphate reductase *archaeon*	35	PHE-7, TYR-12, TYR-22, HIS-23, HIS-24
AF-A0A7C1M999-F1	DUF433 domain-containing protein *Candidatus Pacearchaeota archaeon*	49	TYR-8, PHE-22, TRP-30, TYR-49
AF-A0A7C1R072-F1	3-isopropylmalate dehydratase large subunit *archaeon*	42	PHE-8, HIS-14, TYR-20, TYR-35, TYR-41
AF-A0A7C1R7R2-F1	Rubredoxin *archaeon*	45	TYR-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A7C2AFH7-F1	Ribulose bisphosphate carboxylase large subunit C-terminal domain-containing protein *Thermofilum sp*	39	TYR-19, HIS-23, TRP-33, TYR-35
AF-A0A7C2DAU6-F1	50S ribosomal protein L13 *Candidatus Aenigmatarchaeota archaeon*	31	PHE-17
AF-A0A7C2GI08-F1	Lrp/AsnC family transcriptional regulator *Candidatus Bathyarchaeota archaeon*	42	PHE-5, TYR-35, TYR-38
AF-A0A7C2L465-F1	pyruvate synthase *Nitrososphaeria archaeon*	41	TRP-8, HIS-9, TYR-33, HIS-36, PHE-37, PHE-40
AF-A0A7C2QK30-F1	HEPN domain-containing protein *Thermofilum sp*	49	HIS-8, TRP-10, HIS-17, PHE-20, TYR-21, TRP-24
AF-A0A7C2QRH1-F1	Sugar phosphate isomerase/epimerase *Thermofilum sp*	50	PHE-27, PHE-32, TRP-33, HIS-35
AF-A0A7C2VVD2-F1	Flavodoxin family protein *Candidatus Bathyarchaeota archaeon*	43	PHE-4, TYR-42
AF-A0A7C3M2E6-F1	Histone *Candidatus Aenigmatarchaeota archaeon*	40	TYR-5, HIS-22, HIS-23
AF-A0A7C3QTI1-F1	3-phosphoglycerate dehydrogenase *Desulfurococcaceae archaeon*	44	HIS-10, TYR-26, TYR-29
AF-A0A7C3S3R7-F1	HEPN domain-containing protein *Candidatus Bathyarchaeota archaeon*	40	TRP-8, HIS-18, PHE-27, TYR-28, TRP-30, PHE-33
AF-A0A7C3S9L1-F1	Gfo/Idh/MocA-like oxidoreductase N-terminal domain-containing protein *Candidatus Bathyarchaeota archaeon*	48	TRP-6, HIS-7
AF-A0A7C3SKL2-F1	PIN domain-containing protein *Thermofilum pendens*	47	TYR-19, TYR-31, PHE-33, HIS-36, PHE-37
AF-A0A7C3SL62-F1	Peptidase M28 *Thermofilum pendens*	39	HIS-2, HIS-4, HIS-6, PHE-37
AF-A0A7C3SRL8-F1	Gfo/Idh/MocA family oxidoreductase *Candidatus Bathyarchaeota archaeon*	37	PHE-12, HIS-18, PHE-22
AF-A0A7C3TX69-F1	Nucleoside-diphosphate kinase *Hadesarchaea archaeon*	33	PHE-6

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7C3V221-F1	Phosphoribosylformylglycinamidine synthase *Micrarchaeota archaeon*	32	PHE-12, TYR-22, PHE-24
AF-A0A7C3V662-F1	Peroxiredoxin *Candidatus Bathyarchaeota archaeon*	47	TYR-5, PHE-24, PHE-34, PHE-46
AF-A0A7C3V6M1-F1	ATP-binding cassette domain-containing protein *Candidatus Bathyarchaeota archaeon*	40	PHE-10, TYR-11, TRP-18
AF-A0A7C3VBE0-F1	Translin *Archaeoglobus fulgidus*	42	HIS-30, HIS-39
AF-A0A7C3VLX4-F1	Acetyl-CoA C-acyltransferase *Archaeoglobus fulgidus*	42	PHE-15, TYR-25
AF-A0A7C3WCV6-F1	TNase-like domain-containing protein *Candidatus Aenigmatarchaeota archaeon*	41	TYR-19, TYR-20, PHE-23, TRP-38
AF-A0A7C3YB50-F1	Methionine–tRNA ligase *Candidatus Aenigmatarchaeota archaeon*	42	HIS-5
AF-A0A7C4BA98-F1	Methionine adenosyltransferase *Thermofilum pendens*	46	TYR-12, TYR-18, HIS-30, TYR-33
AF-A0A7C4BAF9-F1	Glutamine-hydrolyzing GMP synthase subunit GuaA *Thermofilum pendens*	38	PHE-3, PHE-8
AF-A0A7C4BEL2-F1	UDP-glucuronate decarboxylase *Hadesarchaea archaeon*	49	PHE-3, PHE-11, HIS-15, PHE-30, TRP-44
AF-A0A7C4C6T0-F1	Nitroreductase domain-containing protein *Candidatus Bathyarchaeota archaeon*	49	TYR-16, TRP-45, HIS-46, PHE-47
AF-A0A7C4C740-F1	TIGR00289 family protein *Candidatus Bathyarchaeota archaeon*	42	TYR-2, PHE-5, TYR-7, PHE-11, PHE-12, TRP-25
AF-A0A7C4CKX8-F1	HEPN domain-containing protein *Nitrososphaeria archaeon*	46	TYR-14, PHE-19, TYR-28, PHE-34, HIS-44
AF-A0A7C4CNR5-F1	Uncharacterized protein *Candidatus Bathyarchaeota archaeon*	26	TYR-6
AF-A0A7C4CNU0-F1	Thiolase domain-containing protein *Candidatus Bathyarchaeota archaeon*	42	PHE-15, HIS-18, TRP-19, TRP-28
AF-A0A7C4CQ25-F1	Imidazoleglycerol-phosphate dehydratase *Candidatus Bathyarchaeota archaeon*	47	HIS-4, PHE-8, HIS-14, HIS-15, PHE-21
AF-A0A7C4CRC4-F1	UDP-N-acetyl-D-glucosamine dehydrogenase *Candidatus Bathyarchaeota archaeon*	49	TRP-13, TYR-26, PHE-31, PHE-40
AF-A0A7C4CV75-F1	Homoaconitate hydratase *Candidatus Bathyarchaeota archaeon*	40	HIS-10, PHE-25
AF-A0A7C4CVP5-F1	Nitroreductase family protein *Nitrososphaeria archaeon*	39	PHE-16, TRP-35
AF-A0A7C4D0G5-F1	acylphosphatase *Candidatus Methanomethylicus sp*	41	PHE-16, TYR-18, PHE-19, TYR-21, TYR-32
AF-A0A7C4D0I2-F1	Phosphoribosylanthranilate isomerase *Ignisphaera aggregans*	45	PHE-26, TYR-38, PHE-43
AF-A0A7C4D0Z0-F1	Acetyl CoA synthetase *Ignisphaera aggregans*	50	TYR-7, PHE-9, HIS-28, TYR-39, TYR-44
AF-A0A7C4D2B3-F1	GTPase *Candidatus Methanomethylicus sp*	38	PHE-19, PHE-21, TRP-24, TYR-30, TYR-34
AF-A0A7C4D2X1-F1	Biotin protein ligase C-terminal domain-containing protein *Thermofilum pendens*	43	HIS-10, PHE-34, TYR-35, HIS-41
AF-A0A7C4D3B7-F1	Nucleotidyltransferase domain-containing protein *Candidatus Bathyarchaeota archaeon*	42	TYR-7, TYR-30, PHE-33
AF-A0A7C4D481-F1	inorganic diphosphatase *Candidatus Bathyarchaeota archaeon*	44	PHE-5, PHE-6, TYR-9, TRP-17, PHE-20, TRP-23, TYR-35, TYR-40
AF-A0A7C4D4U5-F1	Peroxiredoxin *Candidatus Bathyarchaeota archaeon*	39	PHE-12, TRP-26, PHE-31, PHE-32, PHE-34, TRP-35
AF-A0A7C4DB51-F1	Cupin domain-containing protein *Candidatus Bathyarchaeota archaeon*	42	PHE-7, TYR-16, HIS-23, TYR-25, PHE-35
AF-A0A7C4DVK1-F1	GH10 domain-containing protein *Candidatus Bathyarchaeota archaeon*	48	TRP-2, PHE-4, TYR-8, TRP-10, TYR-14, PHE-15, TYR-18, PHE-24, TYR-28, TYR-34, TYR-35, TYR-36, PHE-41, HIS-44
AF-A0A7C4DZ89-F1	tRNA(Ile2) 2-agmatinylcytidine synthetase *Caldiarchaeum subterraneum*	43	TYR-19, HIS-36

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7C4F165-F1	Shikimate dehydrogenase *Hadesarchaea archaeon*	46	TRP-23
AF-A0A7C4FBN7-F1	DegT/DnrJ/EryC1/StrS aminotransferase family protein *Candidatus Bathyarchaeota archaeon*	48	TYR-33, PHE-35, PHE-39, TYR-42, HIS-47
AF-A0A7C4FEM6-F1	Phosphotransferase *Hadesarchaea archaeon*	38	PHE-5, HIS-8, HIS-10, PHE-12, TYR-13
AF-A0A7C4GES3-F1	Uncharacterized protein *Candidatus Aenigmatarchaeota archaeon*	35	PHE-2, PHE-23, PHE-29
AF-A0A7C4GGW0-F1	FAD-binding protein *Thermoproteota archaeon*	36	TRP-5, PHE-22
AF-A0A7C4GN29-F1	FAD-binding domain-containing protein *Thermoproteota archaeon*	32	TRP-2, TYR-18, TYR-25
AF-A0A7C4GR21-F1	Fumarate hydratase *Candidatus Bathyarchaeota archaeon*	48	TYR-4, TYR-5, TYR-25, HIS-38, TYR-44
AF-A0A7C4GSI3-F1	Ribose 1,5-bisphosphate isomerase *Nitrososphaeria archaeon*	44	TRP-19, TYR-41, TYR-42
AF-A0A7C4GSQ4-F1	Molybdenum cofactor biosynthesis protein *Candidatus Bathyarchaeota archaeon*	29	HIS-22, HIS-26
AF-A0A7C4GW05-F1	Pyruvoyl-dependent arginine decarboxylase *Candidatus Bathyarchaeota archaeon*	33	TYR-5, PHE-6, PHE-7, HIS-15, PHE-22, HIS-33
AF-A0A7C4H0M2-F1	acylphosphatase *Candidatus Bathyarchaeota archaeon*	49	TYR-17, TYR-33
AF-A0A7C4H515-F1	ATP-cone domain-containing protein *Candidatus Bathyarchaeota archaeon*	33	PHE-16, HIS-26
AF-A0A7C4H7N1-F1	Ferritin *Candidatus Bathyarchaeota archaeon*	37	PHE-16, HIS-26, HIS-27, PHE-29, PHE-30
AF-A0A7C4HAL7-F1	FAD-dependent oxidoreductase *Candidatus Bathyarchaeota archaeon*	41	TYR-5
AF-A0A7C4HDC8-F1	Flavodoxin family protein *Candidatus Bathyarchaeota archaeon*	38	TYR-8, PHE-20
AF-A0A7C4HL71-F1	Phosphopantetheine adenylyltransferase *Candidatus Bathyarchaeota archaeon*	49	PHE-4, PHE-13, HIS-17, HIS-20, PHE-28, PHE-43
AF-A0A7C4HU33-F1	Universal stress protein *Candidatus Bathyarchaeota archaeon*	42	HIS-14, TYR-16, PHE-21, TYR-29, PHE-39
AF-A0A7C4HZJ5-F1	16S rRNA methyltransferase *Candidatus Bathyarchaeota archaeon*	45	TRP-21, HIS-23, PHE-29, HIS-36
AF-A0A7C4JB99-F1	Uncharacterized protein *Hadesarchaea archaeon*	47	TYR-10, TYR-30, HIS-44
AF-A0A7C4JMT3-F1	Isocitrate/isopropylmalate dehydrogenase family protein *Nitrososphaeria archaeon*	47	PHE-7, TYR-22
AF-A0A7C4JNN4-F1	HEPN domain-containing protein *Nitrososphaeria archaeon*	39	TRP-8, HIS-18, HIS-28, TRP-30, PHE-33
AF-A0A7C4KXK7-F1	Cyclophilin TM1367-like domain-containing protein *Candidatus Methanomethylicus sp*	48	PHE-7, PHE-35, TRP-42, TYR-47, PHE-48
AF-A0A7C4M5D4-F1	DNA replication and repair protein RecF *Candidatus Bathyarchaeota archaeon*	44	HIS-14, PHE-21, PHE-25, TYR-41
AF-A0A7C4M8L5-F1	N-acetyl-gamma-glutamyl-phosphate reductase *Nitrososphaeria archaeon*	45	PHE-7
AF-A0A7C4M8N8-F1	glucose-1-phosphate thymidylyltransferase *Candidatus Bathyarchaeota archaeon*	50	HIS-7, PHE-19, TYR-36
AF-A0A7C4MCQ5-F1	ATP-binding cassette domain-containing protein *Candidatus Bathyarchaeota archaeon*	42	TYR-11, TRP-18
AF-A0A7C4MCW4-F1	Sugar ABC transporter ATP-binding protein *Candidatus Bathyarchaeota archaeon*	39	PHE-13, HIS-21, PHE-24, PHE-31
AF-A0A7C4MFI0-F1	dTDP-glucose 4,6-dehydratase *Candidatus Bathyarchaeota archaeon*	40	HIS-3, PHE-5, TRP-15, TYR-16, TRP-21, TRP-22, TRP-23, HIS-33, TRP-37
AF-A0A7C4NC93-F1	Homoserine dehydrogenase *Hadesarchaea archaeon*	48	PHE-8, PHE-19, HIS-29, PHE-32
AF-A0A7C4NEZ2-F1	Ribose-phosphate pyrophosphokinase *Candidatus Bathyarchaeota archaeon*	49	TYR-23, TYR-30, PHE-33, TYR-39

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7C4NIQ4-F1	Transcription regulator PadR N-terminal domain-containing protein *Candidatus Bathyarchaeota archaeon*	50	TYR-7, HIS-9, TYR-22, PHE-29, PHE-30, PHE-43
AF-A0A7C4NIY2-F1	HEPN domain-containing protein *Candidatus Bathyarchaeota archaeon*	38	PHE-16, TYR-19, PHE-29, PHE-31, PHE-34
AF-A0A7C4NQE1-F1	Thiolase domain-containing protein *Nitrososphaeria archaeon*	40	TYR-19, HIS-24, HIS-31, TYR-32, TYR-34
AF-A0A7C4PVJ1-F1	NAD(P)-dependent glycerol-1-phosphate dehydrogenase *Candidatus Methanomethylicus sp*	41	HIS-10, TYR-17
AF-A0A7C4PZ72-F1	Uncharacterized protein *Candidatus Methanomethylicus sp*	42	PHE-25, TRP-31
AF-A0A7C4Q0N0-F1	Glycosyltransferase *Candidatus Methanomethylicus sp*	42	TYR-10
AF-A0A7C4R5B6-F1	Aldo/keto reductase *Candidatus Bathyarchaeota archaeon*	42	TYR-3, PHE-23
AF-A0A7C4R964-F1	CoA transferase *Nitrososphaeria archaeon*	46	PHE-13, PHE-21
AF-A0A7C4RCS8-F1	CoA transferase *Candidatus Bathyarchaeota archaeon*	42	PHE-14, HIS-16, HIS-22
AF-A0A7C4S1G2-F1	DNA cytosine methyltransferase *Candidatus Bathyarchaeota archaeon*	50	PHE-9, PHE-15, PHE-19, PHE-24, PHE-40, PHE-44
AF-A0A7C4S653-F1	HEPN domain-containing protein *Geoglobus ahangari*	48	PHE-15, TYR-28, PHE-33, HIS-34
AF-A0A7C4S8S6-F1	SDR family oxidoreductase *Nitrososphaeria archaeon*	44	HIS-3, PHE-9, PHE-21, TYR-29, HIS-33, TYR-34, TYR-41
AF-A0A7C4THQ3-F1	peptidyl-tRNA hydrolase *Acidilobales archaeon*	45	TYR-4, HIS-27, TRP-43
AF-A0A7C4TUE3-F1	Methionine–tRNA ligase *Candidatus Bathyarchaeota archaeon*	39	PHE-4, HIS-39
AF-A0A7C4VDL7-F1	Diphthine synthase *Candidatus Woesearchaeota archaeon*	35	TYR-3, HIS-27, TYR-30, TYR-34
AF-A0A7C4VJN0-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Candidatus Bathyarchaeota archaeon*	39	TYR-8, PHE-39
AF-A0A7C4VQX9-F1	Branched-chain amino acid aminotransferase *Candidatus Bathyarchaeota archaeon*	48	PHE-33, TYR-42, TYR-47
AF-A0A7C4W5I9-F1	High-affinity branched-chain amino acid ABC transporter ATP-binding protein LivG *Nitrososphaeria archaeon*	36	TYR-13, TYR-14, TYR-21, PHE-25
AF-A0A7C4W7X4-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Candidatus Bathyarchaeota archaeon*	39	TYR-8
AF-A0A7C4WIF3-F1	TIGR00266 family protein *Candidatus Bathyarchaeota archaeon*	42	TYR-3, TYR-7, TYR-11, TYR-33, HIS-41
AF-A0A7C4XD95-F1	S58 family peptidase *Methanobacteriota archaeon*	50	PHE-2, PHE-18, TYR-46
AF-A0A7C4XSG8-F1	SDR family NAD(P)-dependent oxidoreductase *Acidilobales archaeon*	38	PHE-25
AF-A0A7C4ZS34-F1	Proline racemase *Candidatus Bathyarchaeota archaeon*	47	PHE-5, TYR-13, TYR-25, HIS-31, HIS-42, PHE-44
AF-A0A7C5A9M3-F1	Aspartate–tRNA(Asn) ligase *Candidatus Woesearchaeota archaeon*	49	TRP-22, HIS-24, PHE-34, PHE-37
AF-A0A7C5BHP9-F1	nucleoside-diphosphate kinase *Methanobacteriota archaeon*	41	PHE-25
AF-A0A7C5FAR6-F1	Rubredoxin *Methanobacteriota archaeon*	35	TRP-3, TYR-10, HIS-12, PHE-31
AF-A0A7C5IX38-F1	TATA box-binding protein *Candidatus Bathyarchaeota archaeon*	34	TYR-15, HIS-19, HIS-22, PHE-32, TYR-33
AF-A0A7C5K1Z9-F1	Molybdenum ABC transporter ATP-binding protein *Candidatus Woesearchaeota archaeon*	38	PHE-13, PHE-26

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7C5N6U4-F1	Saccharopine dehydrogenase *Candidatus Bathyarchaeota archaeon*	46	HIS-25
AF-A0A7C5T4J2-F1	superoxide dismutase *Desulfurococcaceae archaeon*	49	TYR-5, TYR-12, TYR-14, TYR-20, HIS-29, TYR-30, HIS-33, HIS-34, TYR-37
AF-A0A7C5T765-F1	ParA family protein *Pyrobaculum sp*	35	PHE-6, HIS-21, HIS-24, PHE-26
AF-A0A7C5T7W6-F1	FAD-dependent oxidoreductase *Pyrobaculum sp*	35	PHE-14, PHE-17, TYR-18
AF-A0A7C5TFR7-F1	CopG family transcriptional regulator *Vulcanisaeta sp*	40	TYR-11, TYR-18, TYR-24
AF-A0A7C5TK25-F1	NAD-dependent dehydratase *Sulfolobales archaeon*	49	TRP-14, HIS-25, TYR-40, PHE-48
AF-A0A7C5U1D8-F1	DegT/DnrJ/EryC1/StrS aminotransferase family protein *Candidatus Bathyarchaeota archaeon*	46	PHE-5, TYR-9, TRP-26, HIS-31, PHE-36, PHE-40, TYR-43
AF-A0A7C5U2B7-F1	Transcriptional regulator *Candidatus Bathyarchaeota archaeon*	44	TYR-11, TRP-13, PHE-16, PHE-17, TRP-20, PHE-30, PHE-37, TRP-42
AF-A0A7C5U5Q9-F1	Quinone oxidoreductase *Caldiarchaeum subterraneum*	38	TYR-17
AF-A0A7C5U7X1-F1	PIN domain-containing protein *Acidilobales archaeon*	30	HIS-7, PHE-16, PHE-20, TRP-25
AF-A0A7C5VQ90-F1	tRNA (cytidine(56)-2'-O)-methyltransferase *Nitrososphaeria archaeon*	36	HIS-11, HIS-22
AF-A0A7C5X1V8-F1	Stationary phase survival protein SurE *Acidilobales archaeon*	34	TYR-14, TYR-22, TYR-32
AF-A0A7C5X2V4-F1	D-glycerate dehydrogenase *Desulfurococcaceae archaeon*	48	PHE-6, PHE-12, TYR-22, PHE-23, TRP-28, TYR-31, HIS-32, HIS-33, TYR-36
AF-A0A7C5XHQ0-F1	Acyl-CoA thioesterase *Vulcanisaeta sp*	43	PHE-18, PHE-27, TRP-34
AF-A0A7C5XHS9-F1	Fumarylacetoacetate hydrolase *Vulcanisaeta sp*	41	HIS-18
AF-A0A7C5XJ97-F1	phosphoribosyl-ATP diphosphatase *Vulcanisaeta sp*	39	TRP-3, TYR-13, HIS-14, HIS-37
AF-A0A7C5XNL1-F1	peptidyl-tRNA hydrolase *Ignisphaera aggregans*	50	TYR-11, HIS-38
AF-A0A7C5XTQ5-F1	Precorrin-6y C5,15-methyltransferase (Decarboxylating) subunit CbiE *Sulfolobales archaeon*	50	TYR-4, TRP-33, PHE-40, PHE-43, PHE-44
AF-A0A7C5Y238-F1	ABC transporter ATP-binding protein *Acidilobales archaeon*	40	PHE-11, PHE-22, TYR-32
AF-A0A7C5Y7Y0-F1	Uncharacterized protein *Caldiarchaeum subterraneum*	39	PHE-5, PHE-12, PHE-15, TYR-26, TYR-33, HIS-37
AF-A0A7C5YCQ1-F1	SDR family NAD(P)-dependent oxidoreductase *Acidilobales archaeon*	40	PHE-28
AF-A0A7C5YSG7-F1	ABC transporter ATP-binding protein *Ignisphaera aggregans*	34	TRP-3, TYR-14, PHE-25, HIS-26, PHE-33
AF-A0A7C5ZRH2-F1	NAD-dependent epimerase *Candidatus Bathyarchaeota archaeon*	36	HIS-2, TYR-11, TYR-13, HIS-15, PHE-20
AF-A0A7C5ZT79-F1	Flagellar accessory protein FlaH *Candidatus Bathyarchaeota archaeon*	47	HIS-21, TYR-31, PHE-38
AF-A0A7C6AIW3-F1	Pyridoxal 5'-phosphate synthase lyase subunit PdxS *Nitrososphaeria archaeon*	46	PHE-19
AF-A0A7C6DX77-F1	Aspartyl protease *Candidatus Bathyarchaeota archaeon*	41	TYR-5, PHE-21, TYR-29, TRP-37
AF-A0A7C6E4A0-F1	Type II toxin-antitoxin system HicB family antitoxin *Candidatus Bathyarchaeota archaeon*	41	PHE-4, TYR-17
AF-A0A7C6FRR6-F1	Pyruvate ferredoxin oxidoreductase *Methanobacterium sp*	41	PHE-1, PHE-21, PHE-37
AF-A0A7C7CNG0-F1	N-acetyltransferase *Candidatus Poseidoniales archaeon*	40	PHE-16, TYR-29, PHE-38

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7C7GE12-F1	N-acetyltransferase *Candidatus Poseidoniales archaeon*	46	TYR-37
AF-A0A7C7GIG1-F1	Glycerol-3-phosphate dehydrogenase *Candidatus Poseidoniales archaeon*	34	TRP-12, TRP-20, HIS-25, TRP-30
AF-A0A7C7MPZ4-F1	7-carboxy-7-deazaguanine synthase *Candidatus Poseidoniales archaeon*	32	PHE-2, TYR-3, TYR-10, HIS-11, PHE-18, PHE-21, TRP-30
AF-A0A7C7PIY6-F1	NADP transhydrogenase beta-like domain-containing protein *Candidatus Poseidoniales archaeon*	46	TRP-7, TYR-11, PHE-18
AF-A0A7C7PPS5-F1	ATP-binding cassette domain-containing protein *Candidatus Poseidoniales archaeon*	50	PHE-14, TYR-15
AF-A0A7C7PU94-F1	Phosphoribosylamine–glycine ligase *Candidatus Poseidoniales archaeon*	40	TYR-18, HIS-30, HIS-31, TYR-37
AF-A0A7C7TZQ8-F1	Histone H1 *Candidatus Poseidoniales archaeon*	50	HIS-15
AF-A0A7C7Y7G9-F1	Glycerophosphodiester phosphodiesterase *Candidatus Poseidoniales archaeon*	38	TYR-2, HIS-15, PHE-20, PHE-25, HIS-38
AF-A0A7G9Y4P4-F1	Uncharacterized protein *Candidatus Methanogaster sp. ANME-2c ERB4*	48	PHE-20, PHE-30, TYR-35
AF-A0A7G9YHQ3-F1	Desulfoferrodoxin N-terminal domain-containing protein *Candidatus Methanogaster sp. ANME-2c ERB4*	37	TYR-8
AF-A0A7G9YP81-F1	Rubredoxin *Candidatus Methanogaster sp. ANME-2c ERB4*	46	TYR-4, TYR-6, PHE-23, TRP-30, PHE-41, PHE-42
AF-A0A7G9YP82-F1	Desulfoferrodoxin N-terminal domain-containing protein *Candidatus Methanogaster sp. ANME-2c ERB4*	37	TYR-8
AF-A0A7G9YXX7-F1	Rubredoxin *Candidatus Methanophagaceae archaeon ANME-1 ERB6*	45	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A7G9Z2P5-F1	Uncharacterized protein *Candidatus Methanophaga sp. ANME-1 ERB7*	48	TYR-8
AF-A0A7G9Z8V8-F1	UDP-N-acetyl-D-mannosamine dehydrogenase *Candidatus Methanophaga sp. ANME-1 ERB7*	47	TRP-13, TYR-21, TYR-42, TRP-47
AF-A0A7G9ZD30-F1	Rubredoxin *Candidatus Methanophaga sp. ANME-1 ERB7*	45	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A7G9ZD32-F1	Rubredoxin *Candidatus Methanophaga sp. ANME-1 ERB7*	45	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A7H1KP24-F1	Uncharacterized protein *uncultured Methanosarcinales archaeon*	48	PHE-20, PHE-30, TYR-35
AF-A0A7J2H756-F1	Aldo/keto reductase *Thermofilaceae archaeon*	49	TYR-3, PHE-18, PHE-45
AF-A0A7J2IPL3-F1	50S ribosomal protein L4 *Candidatus Bathyarchaeota archaeon*	45	TYR-6, PHE-22, PHE-26
AF-A0A7J2IQR4-F1	Formylmethanofuran–tetrahydromethanopterin N-formyltransferase *Candidatus Bathyarchaeota archaeon*	46	PHE-2, TRP-5, TRP-18, PHE-29
AF-A0A7J2ISG1-F1	HEPN domain-containing protein *Candidatus Bathyarchaeota archaeon*	41	PHE-9, PHE-27, TYR-35, TYR-36, PHE-39
AF-A0A7J2LST6-F1	Carbamoyltransferase *Candidatus Bathyarchaeota archaeon*	38	TYR-1, HIS-4, PHE-19, TYR-26, TYR-32
AF-A0A7J2MF60-F1	MBL fold metallo-hydrolase *Methanosarcinales archaeon*	43	TYR-16, PHE-17, TYR-25, TYR-26, PHE-36, TYR-40
AF-A0A7J2MGK5-F1	N-(5'-phosphoribosyl)anthranilate isomerase *Methanosarcinales archaeon*	32	TYR-19, PHE-30
AF-A0A7J2MGV4-F1	F420-dependent methylenetetrahydromethanopterin dehydrogenase *Methanosarcinales archaeon*	50	HIS-5, PHE-29, HIS-33, TYR-46
AF-A0A7J2MHH4-F1	Ornithine cyclodeaminase family protein *Methanosarcinales archaeon*	44	PHE-12, HIS-27, TYR-29, HIS-40, PHE-44
AF-A0A7J2PVB2-F1	Glutamate dehydrogenase *archaeon*	50	PHE-3, TYR-14, PHE-24, HIS-27, TYR-33, TRP-49
AF-A0A7J2PZK0-F1	Nitroreductase *archaeon*	48	PHE-3, TYR-8, PHE-16, PHE-32, PHE-47

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7J2QZC8-F1	ATP-binding cassette domain-containing protein *archaeon*	44	PHE-4
AF-A0A7J2R6L5-F1	DNA cytosine methyltransferase *archaeon*	40	PHE-17, HIS-22, PHE-25, HIS-36, TRP-39, TYR-40
AF-A0A7J2R8L4-F1	PHP domain-containing protein *archaeon*	40	HIS-6, HIS-8, HIS-38
AF-A0A7J2T743-F1	Type I glyceraldehyde-3-phosphate dehydrogenase *Candidatus Bathyarchaeota archaeon*	37	TRP-13, TYR-14, TRP-18, PHE-20, PHE-37
AF-A0A7J2TJN1-F1	Phosphoribosylglycinamide synthetase N-terminal domain-containing protein *Archaeoglobus fulgidus*	35	HIS-17, PHE-19, TYR-29
AF-A0A7J2TN26-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Candidatus Bathyarchaeota archaeon*	38	TYR-8, PHE-9
AF-A0A7J2TXK4-F1	Rubredoxin *Candidatus Aenigmatarchaeota archaeon*	44	TRP-3, TYR-10, TYR-12, PHE-21, TRP-28, PHE-40
AF-A0A7J2UQV7-F1	Kinase *Hadesarchaea archaeon*	42	HIS-4, PHE-8, PHE-9, PHE-12
AF-A0A7J2V0Z7-F1	PHP domain-containing protein *Acidilobales archaeon*	42	HIS-10, HIS-40
AF-A0A7J2V9U9-F1	HEPN domain-containing protein *Desulfurococcaceae archaeon*	33	TRP-8, PHE-9, PHE-16, HIS-18, TYR-28, TRP-30, PHE-33
AF-A0A7J2VBC6-F1	ABC transporter ATP-binding protein *Desulfurococcaceae archaeon*	44	PHE-3, PHE-14, HIS-22
AF-A0A7J2VTQ0-F1	Phenylacetate–CoA ligase *Candidatus Aenigmatarchaeota archaeon*	42	TYR-22, PHE-24, HIS-25, TYR-27, TYR-28, PHE-32
AF-A0A7J2VWF2-F1	Phenylacetate–CoA ligase *Candidatus Aenigmatarchaeota archaeon*	42	TYR-22, PHE-24, HIS-25, TYR-27, TYR-28, PHE-32
AF-A0A7J2WFC1-F1	SDR family NAD(P)-dependent oxidoreductase *Desulfurococcaceae archaeon*	42	PHE-28
AF-A0A7J2Y502-F1	tRNA (cytidine(56)-2'-O)-methyltransferase *Methanobacteriota archaeon*	37	HIS-9, HIS-20, PHE-28
AF-A0A7J2Y6X9-F1	NAD(P)-dependent oxidoreductase *Methanobacteriota archaeon*	28	PHE-12, HIS-16, TRP-26
AF-A0A7J2YMB9-F1	Bifunctional formaldehyde-activating protein/3-hexulose-6-phosphate synthase *Candidatus Bathyarchaeota archaeon*	45	TYR-2, HIS-17, PHE-33, HIS-44
AF-A0A7J2Z0G0-F1	Methionyl/Leucyl tRNA synthetase domain-containing protein *Candidatus Aenigmatarchaeota archaeon*	47	TYR-14, HIS-20, HIS-23, TYR-28, PHE-34, PHE-37, HIS-43, HIS-47
AF-A0A7J2ZCD7-F1	Adenosine monophosphate-protein transferase *Candidatus Bathyarchaeota archaeon*	45	PHE-19, PHE-36, TYR-42
AF-A0A7J2ZIP3-F1	acylphosphatase *Nitrososphaeria archaeon*	45	HIS-5, PHE-7, PHE-16, PHE-17, TRP-33
AF-A0A7J2ZM04-F1	UDP-glucose/GDP-mannose dehydrogenase N-terminal domain-containing protein *Candidatus Aenigmatarchaeota archaeon*	34	TYR-10, HIS-21, HIS-24
AF-A0A7J3AC04-F1	Ammonium transporter *Candidatus Bathyarchaeota archaeon*	48	PHE-6, PHE-8, PHE-9, TYR-12, HIS-40, TYR-45, HIS-46
AF-A0A7J3ADC8-F1	Daunorubicin ABC transporter ATP-binding protein *Candidatus Bathyarchaeota archaeon*	39	PHE-3, TYR-14, HIS-33, PHE-35
AF-A0A7J3B0X2-F1	EVE domain-containing protein *Thermoproteota archaeon*	26	TYR-3, TRP-4, TRP-13, TRP-22
AF-A0A7J3B843-F1	IMP dehydrogenase/GMP reductase domain-containing protein *Methanobacteriota archaeon*	48	PHE-2, TYR-13, TYR-15, TYR-37, HIS-41
AF-A0A7J3BC48-F1	Arsenate reductase ArsC *Candidatus Bathyarchaeota archaeon*	43	PHE-6, PHE-21, TYR-25, PHE-30

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7J3BD52-F1	ATPase F1/V1/A1 complex alpha/beta subunit N-terminal domain-containing protein *Candidatus Bathyarchaeota archaeon*	42	TYR-23
AF-A0A7J3BEY8-F1	5-(Carboxyamino)imidazole ribonucleotide mutase *Candidatus Bathyarchaeota archaeon*	31	HIS-21
AF-A0A7J3C7I1-F1	DUF521 domain-containing protein *Methanobacteriota archaeon*	36	TYR-2, PHE-19, PHE-33
AF-A0A7J3CJ29-F1	Subtype I-B CRISPR-associated endonuclease Cas1 *Candidatus Aenigmatarchaeota archaeon*	40	TYR-6, PHE-8, TYR-21, PHE-22, TYR-30, PHE-40
AF-A0A7J3D6Q8-F1	Argininosuccinate synthase *Nitrososphaeria archaeon*	42	TYR-3, TYR-10, PHE-16, PHE-24, TRP-28, TYR-36
AF-A0A7J3D8F9-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanomicrobia archaeon*	39	TYR-10
AF-A0A7J3D9G8-F1	3,4-dihydroxy-2-butanone-4-phosphate synthase *Methanomicrobia archaeon*	48	PHE-2, TYR-21
AF-A0A7J3DAY3-F1	Aldehyde ferredoxin oxidoreductase N-terminal domain-containing protein *Methanomicrobia archaeon*	38	TYR-2, TYR-4, TRP-5, PHE-23, PHE-27, TRP-31, PHE-37
AF-A0A7J3DNK5-F1	50S ribosomal protein L34 *archaeon*	44	TYR-5, HIS-16, PHE-18, TRP-40
AF-A0A7J3EGB7-F1	Aminopeptidase *Candidatus Bathyarchaeota archaeon*	35	TYR-4, TRP-16, TYR-25, TYR-28
AF-A0A7J3ELT1-F1	O-acetyl-ADP-ribose deacetylase *Desulfurococcaceae archaeon*	50	TYR-4, TYR-33, HIS-36
AF-A0A7J3EMJ0-F1	YjbQ family protein *Desulfurococcaceae archaeon*	48	HIS-1, PHE-12, TRP-25
AF-A0A7J3F9X2-F1	Formylmethanofuran–tetrahydromethanopterin N-formyltransferase *Candidatus Bathyarchaeota archaeon*	39	TYR-23, HIS-30, PHE-33
AF-A0A7J3FAQ0-F1	Sugar phosphate isomerase/epimerase *Candidatus Bathyarchaeota archaeon*	47	TRP-5, TYR-8, TYR-17, TYR-19, HIS-25, PHE-29, PHE-30, HIS-42
AF-A0A7J3FC68-F1	4-alpha-glucanotransferase *Hadesarchaea archaeon*	45	HIS-9, PHE-10, HIS-17, TRP-25, PHE-29, TYR-40, TRP-41, PHE-43
AF-A0A7J3FC71-F1	6-phosphofructokinase *Candidatus Bathyarchaeota archaeon*	50	TYR-29, PHE-31, TRP-40
AF-A0A7J3FCW3-F1	FAD-binding protein *Hadesarchaea archaeon*	38	TYR-20
AF-A0A7J3GV54-F1	Aspartate-semialdehyde dehydrogenase *Nitrososphaeria archaeon*	50	HIS-6, PHE-21
AF-A0A7J3HB40-F1	ATP-binding cassette domain-containing protein *Desulfurococcaceae archaeon*	50	TYR-11, PHE-23
AF-A0A7J3HP86-F1	ATP-binding cassette domain-containing protein *Nitrososphaeria archaeon*	48	TYR-14, HIS-18, PHE-21
AF-A0A7J3HPJ8-F1	Shikimate dehydrogenase substrate binding N-terminal domain-containing protein *Nitrososphaeria archaeon*	35	PHE-3, HIS-9, HIS-13, HIS-20, TRP-25, TYR-27, HIS-31, TYR-33
AF-A0A7J3I0Z5-F1	Tetrahydromethanopterin S-methyltransferase subunit H *Candidatus Bathyarchaeota archaeon*	34	PHE-5, PHE-12
AF-A0A7J3I3M4-F1	MOFRL domain-containing protein *Candidatus Bathyarchaeota archaeon*	39	TYR-3, HIS-7, PHE-12, PHE-13
AF-A0A7J3I437-F1	FAD-binding protein *Candidatus Bathyarchaeota archaeon*	34	PHE-2
AF-A0A7J3I6Q7-F1	Glycerate kinase *Ignisphaera aggregans*	46	TYR-2, TYR-11, TYR-18, TYR-20, PHE-21, TYR-32, PHE-41
AF-A0A7J3IA32-F1	Acetyl-CoA synthetase *Ignisphaera aggregans*	42	PHE-24, TYR-26
AF-A0A7J3IFT0-F1	Chorismate-utilising enzyme C-terminal domain-containing protein *Candidatus Bathyarchaeota archaeon*	40	HIS-5, TRP-22, PHE-27

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7J3IY93-F1	Aldehyde ferredoxin oxidoreductase N-terminal domain-containing protein *Candidatus Bathyarchaeota archaeon*	49	TYR-2, TYR-4, HIS-10, HIS-19, TYR-27, TYR-31, TRP-42, TYR-45
AF-A0A7J3J0Q3-F1	Bifunctional hydroxymethylpyrimidine kinase/phosphomethylpyrimidine kinase *Candidatus Bathyarchaeota archaeon*	36	PHE-28, HIS-34
AF-A0A7J3J2N7-F1	DUF424 family protein *Candidatus Bathyarchaeota archaeon*	38	TYR-18, HIS-20, HIS-32
AF-A0A7J3J3L3-F1	3-hydroxyacyl-CoA dehydrogenase NAD binding domain-containing protein *Candidatus Bathyarchaeota archaeon*	45	HIS-3, HIS-32, PHE-33, PHE-34
AF-A0A7J3JCS0-F1	DUF4152 domain-containing protein *Candidatus Bathyarchaeota archaeon*	39	TYR-16, PHE-17, PHE-22, TYR-34
AF-A0A7J3KAV9-F1	HEPN domain-containing protein *Candidatus Bathyarchaeota archaeon*	49	TRP-8, TYR-16, HIS-18, TYR-28, TRP-30, PHE-33, HIS-36, TYR-47
AF-A0A7J3LWS6-F1	3-isopropylmalate dehydratase small subunit *Candidatus Bathyarchaeota archaeon*	35	HIS-14, TYR-25
AF-A0A7J3LXD3-F1	LLM class flavin-dependent oxidoreductase *Candidatus Bathyarchaeota archaeon*	48	PHE-3, PHE-7, PHE-13, TRP-14, TYR-19, PHE-28, TYR-30, TRP-32, HIS-36, TYR-37, PHE-43
AF-A0A7J3LYC8-F1	Ribbon-helix-helix protein, CopG family *Candidatus Bathyarchaeota archaeon*	35	TYR-7, TYR-12, TYR-25, PHE-34
AF-A0A7J3M1M9-F1	YjbQ family protein *Archaeoglobus fulgidus*	33	PHE-2, TRP-12, PHE-17, PHE-20, TYR-29
AF-A0A7J3M2B2-F1	30S ribosomal protein S27e *Archaeoglobus fulgidus*	39	PHE-4, HIS-6, TYR-38
AF-A0A7J3M345-F1	CRISPR-associated endoribonuclease Cas6 *Archaeoglobus fulgidus*	43	TRP-4, HIS-5, PHE-12, TYR-25, PHE-29, PHE-37
AF-A0A7J3M4E2-F1	Methenyltetrahydromethanopterin cyclohydrolase *Archaeoglobus fulgidus*	42	TYR-1, PHE-8, TYR-24, HIS-26, PHE-35, TYR-38
AF-A0A7J3M654-F1	NAD(P)-dependent oxidoreductase *Candidatus Bathyarchaeota archaeon*	35	PHE-11, HIS-19, TYR-25, PHE-30
AF-A0A7J3MGX5-F1	Oligopeptide ABC transporter ATP-binding protein *Candidatus Bathyarchaeota archaeon*	47	PHE-6, TRP-14, PHE-15, PHE-22, TYR-32, PHE-41
AF-A0A7J3MYD0-F1	CMP/dCMP-type deaminase domain-containing protein *Ignisphaera aggregans*	46	TYR-12, TYR-22
AF-A0A7J3MZ99-F1	3-isopropylmalate dehydratase large subunit *Ignisphaera aggregans*	45	PHE-10, TYR-22
AF-A0A7J3P4N5-F1	superoxide dismutase *Nitrososphaeria archaeon*	37	TYR-4, TYR-11, TYR-13, TYR-19, HIS-28, HIS-29, HIS-32, HIS-33, TYR-36
AF-A0A7J3PA89-F1	Thiolase domain-containing protein *Hadesarchaea archaeon*	47	HIS-29, HIS-41, PHE-43
AF-A0A7J3QFR6-F1	5-(Carboxyamino)imidazole ribonucleotide mutase *Hadesarchaea archaeon*	46	PHE-16, PHE-27, PHE-31, HIS-39
AF-A0A7J3QSA6-F1	Glycine cleavage system protein H *Candidatus Bathyarchaeota archaeon*	50	TYR-12, TRP-16, PHE-35, TRP-41, HIS-42
AF-A0A7J3QSZ4-F1	Uncharacterized protein *Candidatus Bathyarchaeota archaeon*	48	TYR-6, TYR-37, PHE-42, TYR-44
AF-A0A7J3QT08-F1	tRNA pseudouridine(13) synthase TruD *Candidatus Bathyarchaeota archaeon*	41	TYR-15, PHE-33
AF-A0A7J3QT34-F1	FAD-dependent oxidoreductase *Candidatus Bathyarchaeota archaeon*	49	PHE-3
AF-A0A7J3QTT3-F1	DNA helicase UvrD *Candidatus Bathyarchaeota archaeon*	46	HIS-8, HIS-10, TYR-13, HIS-19, PHE-29, PHE-43, HIS-45
AF-A0A7J3QYA8-F1	Mevalonate kinase *Candidatus Bathyarchaeota archaeon*	34	PHE-15, HIS-18, PHE-19, TYR-22
AF-A0A7J3R097-F1	Ornithine cyclodeaminase family protein *Candidatus Bathyarchaeota archaeon*	37	PHE-12, HIS-19, TYR-27, HIS-29, TYR-30, HIS-33
AF-A0A7J3S713-F1	LSM domain protein *Methanobacteriaceae archaeon*	49	TYR-23, PHE-36, PHE-44

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7J3SZL7-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Candidatus Bathyarchaeota archaeon*	39	TYR-8
AF-A0A7J3T0Q9-F1	Pseudouridine synthase *Candidatus Bathyarchaeota archaeon*	47	HIS-3, PHE-33
AF-A0A7J3TGH2-F1	Nicotinamide-nucleotide adenylyltransferase *Thermoplasmatales archaeon*	36	PHE-5, PHE-10, PHE-13, HIS-14, HIS-17
AF-A0A7J3TX92-F1	Gfo/Idh/MocA-like oxidoreductase N-terminal domain-containing protein *Candidatus Bathyarchaeota archaeon*	49	PHE-14, TRP-15, HIS-19, TYR-23
AF-A0A7J3TXK2-F1	Imidazole glycerol phosphate synthase subunit HisF *Candidatus Bathyarchaeota archaeon*	45	HIS-3, PHE-22, HIS-23, TYR-24, TYR-27, TYR-35, HIS-39
AF-A0A7J3UYK2-F1	HEPN domain-containing protein *Candidatus Methanosuratincola petrocarbonis*	38	TRP-7, TRP-8, TYR-22, PHE-23, PHE-27, PHE-33
AF-A0A7J3W5A3-F1	Rubredoxin *Candidatus Aenigmatarchaeota archaeon*	44	TRP-3, TYR-10, TYR-12, PHE-21, TRP-28, PHE-40
AF-A0A7J3W5T9-F1	Arsenate reductase ArsC *Candidatus Aenigmatarchaeota archaeon*	40	PHE-7, HIS-11, PHE-22, TYR-25, TYR-27, PHE-31, TYR-34
AF-A0A7J3W6K5-F1	Fumarate hydratase *Candidatus Bathyarchaeota archaeon*	38	PHE-4, PHE-6, PHE-38
AF-A0A7J3W6T1-F1	Lrp/AsnC family transcriptional regulator *Candidatus Bathyarchaeota archaeon*	33	PHE-12, PHE-30, TYR-33
AF-A0A7J3W7Q6-F1	Macrolide ABC transporter ATP-binding protein *Candidatus Bathyarchaeota archaeon*	49	TYR-14, HIS-22, TYR-36
AF-A0A7J3W909-F1	Nucleotidyl transferase domain-containing protein *Candidatus Bathyarchaeota archaeon*	41	PHE-17
AF-A0A7J3WCY0-F1	Universal stress protein *Caldiarchaeum subterraneum*	39	PHE-2, TYR-14, PHE-35, HIS-37
AF-A0A7J3WZ86-F1	HEPN domain-containing protein *Nitrososphaeria archaeon*	44	PHE-4, TRP-10, PHE-25, PHE-27, TYR-30, TRP-32, PHE-35
AF-A0A7J3XGR4-F1	Peptide chain release factor 1 *Acidilobales archaeon*	37	TRP-18, HIS-21, TYR-29
AF-A0A7J3XUF5-F1	Energy-converting hydrogenase A, subunit R *Candidatus Bathyarchaeota archaeon*	48	PHE-5, PHE-20, HIS-25, TYR-26, PHE-33, PHE-34, TYR-40
AF-A0A7J3XUT2-F1	Aromatic acid decarboxylase *Candidatus Bathyarchaeota archaeon*	40	HIS-31
AF-A0A7J3ZAE1-F1	Peroxisomal trans-2-enoyl-CoA reductase *Acidilobales archaeon*	45	PHE-22, TYR-30
AF-A0A7J3ZM45-F1	AbrB/MazE/SpoVT family DNA-binding domain-containing protein *Candidatus Bathyarchaeota archaeon*	48	PHE-14, PHE-15, TRP-22
AF-A0A7J3ZQ93-F1	Nitroreductase domain-containing protein *Candidatus Bathyarchaeota archaeon*	39	TYR-6, TRP-14, TYR-22
AF-A0A7J4B0F6-F1	ATP-binding cassette domain-containing protein *Candidatus Bathyarchaeota archaeon*	41	TYR-10, TYR-11, TRP-18, PHE-23
AF-A0A7J4B2F1-F1	Ankyrin repeat domain-containing protein *Micrarchaeota archaeon*	45	TYR-31
AF-A0A7J4BA39-F1	Cobalamin-binding protein *Desulfurococcaceae archaeon*	46	TRP-19, PHE-33
AF-A0A7J4BAQ2-F1	RNA 3'-terminal phosphate cyclase *Candidatus Aenigmatarchaeota archaeon*	41	PHE-14
AF-A0A7J4EKQ9-F1	Recombinase *Methanosarcinales archaeon*	38	TYR-18
AF-A0A7J4FTH4-F1	IS91 family transposase *Methanosarcinales archaeon*	40	HIS-11, PHE-13, PHE-19, HIS-22, HIS-24, PHE-32, PHE-38, HIS-40
AF-A0A7J4FUL6-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanosarcinales archaeon*	38	TYR-8
AF-A0A7J4G0F4-F1	Peroxiredoxin *Candidatus Geothermarchaeota archaeon*	47	PHE-9, TYR-31, PHE-36, PHE-39, TRP-40

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7J4NWE9-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanosarcinales archaeon*	40	TYR-10
AF-A0A7J9SMA8-F1	GNAT family N-acetyltransferase *ANME-2 cluster archaeon*	48	HIS-3, TRP-9, TYR-16, TYR-19, TYR-24, HIS-33, TYR-42, PHE-43
AF-A0A7J9SPI1-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *ANME-2 cluster archaeon*	40	TYR-10
AF-A0A7K3Y6W1-F1	Transposase *Methanomicrobiales archaeon*	31	HIS-3, HIS-7
AF-A0A7K3YIN1-F1	Carbamoyl phosphate synthase small subunit *Methanomicrobiales archaeon*	45	PHE-18, PHE-30, TYR-37
AF-A0A7K4GQA7-F1	Rubredoxin *Promethearchaeota archaeon*	44	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-40, PHE-41
AF-A0A7K4NEG8-F1	Ketol-acid reductoisomerase *Marine Group I thaumarchaeote*	44	TRP-5, TYR-25
AF-A0A7K4NF07-F1	IMP dehydrogenase *Marine Group I thaumarchaeote*	49	PHE-9, PHE-18
AF-A0A7L4QJ31-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanomassiliicoccales archaeon*	38	TYR-8
AF-A0A7M3SGS7-F1	isocitrate dehydrogenase (NADP(+)) *Candidatus Poseidoniales archaeon*	46	PHE-1, TYR-27, HIS-40
AF-A0A7M3SKX7-F1	NAD-dependent epimerase/dehydratase family protein *Candidatus Poseidoniales archaeon*	34	PHE-11, HIS-15
AF-A0A7M3W8C7-F1	Peptidase S9 prolyl oligopeptidase catalytic domain-containing protein *Candidatus Poseidoniales archaeon*	49	HIS-4, TYR-17, TRP-21, PHE-39, PHE-47, TYR-48
AF-A0A7M3WGX1-F1	Aminotransferase class V-fold PLP-dependent enzyme *Candidatus Poseidoniales archaeon*	40	HIS-2, HIS-3, PHE-22, TRP-23, TYR-25, PHE-34, HIS-37, TYR-40
AF-A0A7M3WHN5-F1	NAD-dependent epimerase/dehydratase family protein *Candidatus Poseidoniales archaeon*	32	PHE-11, HIS-15
AF-A0A7M3WSQ7-F1	FAD-binding protein *Candidatus Poseidoniales archaeon*	33	HIS-22, HIS-26, TYR-31
AF-A0A7M3WYC8-F1	peptidyl-tRNA hydrolase *Candidatus Poseidoniales archaeon*	42	HIS-10, TRP-30
AF-A0A7M3X120-F1	NAD-dependent epimerase/dehydratase family protein *Candidatus Poseidoniales archaeon*	47	PHE-11, HIS-15, HIS-37, PHE-40, HIS-44
AF-A0A7M3X1A0-F1	6-carboxytetrahydropterin synthase QueD *Candidatus Poseidoniales archaeon*	40	TYR-8, HIS-15, HIS-20, HIS-26, HIS-28, HIS-30, TYR-32, PHE-34
AF-A0A7M3X379-F1	peptidyl-tRNA hydrolase *Candidatus Poseidoniales archaeon*	42	TYR-3, HIS-38
AF-A0A7M3X6H7-F1	nucleoside-diphosphate kinase *Candidatus Poseidoniales archaeon*	37	TYR-5, PHE-25, TYR-37
AF-A0A7M3XQK6-F1	NAD-dependent epimerase/dehydratase family protein *Candidatus Poseidoniales archaeon*	29	PHE-11, HIS-15, TYR-25
AF-A0A7M3XRS6-F1	Fructose-bisphosphatase class II *Candidatus Poseidoniales archaeon*	40	PHE-9, PHE-10, TRP-24
AF-A0A7M3XTS2-F1	CoA carboxyltransferase N-terminal domain-containing protein *Candidatus Poseidoniales archaeon*	49	HIS-12, PHE-32, HIS-41, PHE-46
AF-A0A7M3XVG2-F1	Cystathionine gamma-synthase *Candidatus Poseidoniales archaeon*	46	PHE-2, HIS-6, PHE-23, TRP-37
AF-A0A7M3XVL2-F1	ATP-binding cassette domain-containing protein *Candidatus Poseidoniales archaeon*	48	TYR-15, PHE-27, HIS-30
AF-A0A7M3XYR1-F1	tRNA synthetases class I catalytic domain-containing protein *Candidatus Poseidoniales archaeon*	42	PHE-3, HIS-8, HIS-11, HIS-16, HIS-17, TRP-18, HIS-20, PHE-23, PHE-38, TRP-39
AF-A0A7M3XZU0-F1	FAD synthase *Candidatus Poseidoniales archaeon*	36	PHE-10, HIS-14, HIS-17, HIS-19, TYR-20

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Table 1 continued from the preceding page

AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A7M3YDZ0-F1	Ribose 5-phosphate isomerase A *Candidatus Poseidoniales archaeon*	46	PHE-18, TYR-35
AF-A0A7M3YIS4-F1	DUF2237 family protein *Candidatus Poseidoniales archaeon*	50	TRP-23, TYR-24, HIS-28, HIS-39, PHE-49
AF-A0A7M3YK94-F1	Glycerol-3-phosphate dehydrogenase *Candidatus Poseidoniales archaeon*	47	HIS-3, TRP-12, TRP-20, HIS-25, HIS-26, TRP-30, TYR-37
AF-A0A7M3YPB3-F1	3-oxoacyl-ACP reductase *Candidatus Poseidoniales archaeon*	25	HIS-22
AF-A0A7M3YUC1-F1	Glyceraldehyde-3-phosphate dehydrogenase *Candidatus Poseidoniales archaeon*	50	TYR-4, TYR-5, PHE-6, HIS-10, TRP-33, TYR-47
AF-A0A7M3Z7R6-F1	SDR family oxidoreductase *Candidatus Poseidoniales archaeon*	49	TRP-3, PHE-39, TYR-42, TRP-49
AF-A0A7M3Z7W4-F1	Glutamine-hydrolyzing GMP synthase subunit GuaA *Candidatus Poseidoniales archaeon*	43	PHE-2, PHE-7
AF-A0A7M3ZAB9-F1	Aldehyde dehydrogenase family protein *Candidatus Poseidoniales archaeon*	48	TYR-8, TRP-13
AF-A0A7M3ZI84-F1	3,4-dihydroxy-2-butanone-4-phosphate synthase *Candidatus Poseidoniales archaeon*	40	HIS-15
AF-A0A7M3ZII4-F1	peptide-methionine (R)-S-oxide reductase *Candidatus Poseidoniales archaeon*	38	HIS-8, HIS-11, PHE-13, HIS-20, TYR-25, PHE-34
AF-A0A7M3ZZV4-F1	NAD(P)(+) transhydrogenase (Re/Si-specific) subunit alpha *Candidatus Poseidoniales archaeon*	50	PHE-30, TYR-33, TYR-44, HIS-45, TYR-49
AF-A0A7M4A1E0-F1	30S ribosomal protein S8 *Candidatus Poseidoniales archaeon*	32	PHE-1, HIS-14, HIS-15, TYR-16, TYR-30, TYR-32
AF-A0A7M4A5Q8-F1	Ferredoxin *Candidatus Poseidoniales archaeon*	43	HIS-3, HIS-12, TYR-14, PHE-26, TYR-32
AF-A0A7M4AAC8-F1	Nicotinate dehydrogenase medium molybdopterin subunit *Candidatus Poseidoniales archaeon*	46	TYR-2, TRP-21, HIS-24, HIS-44
AF-A0A7M4AP21-F1	4-demethylwyosine synthase TYW1 *Candidatus Poseidoniales archaeon*	48	TYR-17, HIS-18, HIS-23, HIS-30, TRP-31, TYR-43, PHE-47, TYR-48
AF-A0A811T5J5-F1	Rubredoxin *Candidatus Argoarchaeum ethanivorans*	45	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A811TB18-F1	Rubredoxin *Candidatus Argoarchaeum ethanivorans*	45	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A811TBK2-F1	Uncharacterized protein *Candidatus Argoarchaeum ethanivorans*	38	TYR-36, TRP-37
AF-A0A811TE36-F1	Rubredoxin *Candidatus Argoarchaeum ethanivorans*	45	TRP-4, TYR-11, TYR-13, PHE-22, TRP-29, PHE-41
AF-A0A812A2F1-F1	Nitrogen fixation nifHD region glnB-like protein1 *Candidatus Argoarchaeum ethanivorans*	48	PHE-25, PHE-31, TYR-39
AF-A0A822IIB9-F1	Partial Shikimate dehydrogenase (NADP(+)) *Methanosarcinales archaeon*	46	PHE-4, HIS-13, HIS-20, PHE-24, TYR-33, HIS-34, PHE-36, HIS-44
AF-A0A822IKI2-F1	Uncharacterized protein *Methanosarcinales archaeon*	33	PHE-12
AF-A0A822ILJ2-F1	Partial Carbamoyl-phosphate synthase large chain *Methanosarcinales archaeon*	36	PHE-26, PHE-28
AF-A0A822IN67-F1	3-isopropylmalate dehydratase large subunit *Methanosarcinales archaeon*	48	PHE-22, TYR-29, HIS-33, PHE-44
AF-A0A822IR84-F1	Partial Quinate/shikimate dehydrogenase *Methanosarcinales archaeon*	44	TYR-16, TRP-25
AF-A0A822IUL2-F1	tRNA-binding domain-containing protein *Methanosarcinales archaeon*	42	PHE-10
AF-A0A831V7D1-F1	Acetyl-CoA C-acyltransferase *Thermoproteus sp*	48	HIS-4
AF-A0A831VDP7-F1	GTP-binding protein *Thermoproteus sp*	44	TRP-5
AF-A0A831YJC5-F1	CBS domain-containing protein *Pyrobaculum sp*	44	HIS-30
AF-A0A832DFU9-F1	DUF371 domain-containing protein *Pyrobaculum sp*	47	TYR-8, TYR-47

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A0A832VUI6-F1	Cold-shock protein *Candidatus Poseidoniales archaeon*	28	PHE-10
AF-A0A832Y0B8-F1	Acetyl-CoA acetyltransferase *Candidatus Poseidoniales archaeon*	48	HIS-33, HIS-43
AF-A0A832YEZ8-F1	FAD-dependent oxidoreductase *Candidatus Poseidoniales archaeon*	46	TYR-18
AF-A0A842Q616-F1	5-(Carboxyamino)imidazole ribonucleotide mutase *Nitrososphaeraceae archaeon*	30	TYR-8, PHE-20
AF-A0A842Y7E2-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanosarcinales archaeon*	40	TYR-10
AF-A0A843BFT9-F1	Nucleotide exchange factor GrpE *Nitrososphaerota archaeon*	50	TYR-27
AF-A0A843CAZ7-F1	CBS domain-containing protein *Methanosarcinales archaeon*	48	TYR-34, HIS-44
AF-A0A849PHU7-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanosarcinales archaeon*	37	TYR-8
AF-A0A849PJP1-F1	Nitroreductase family protein *Methanosarcinales archaeon*	43	TYR-16, TRP-35
AF-A0A849PLI1-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Methanosarcinales archaeon*	37	TYR-8
AF-A0A8J7XH87-F1	NFYB/HAP3 family transcription factor subunit *Theionarchaea archaeon*	42	HIS-32
AF-A0A8J8C3Q5-F1	DUF521 domain-containing protein *Theionarchaea archaeon*	48	HIS-2, TYR-33
AF-A0A8J8DHJ8-F1	Adenosine monophosphate-protein transferase *Thermococcus sp. MAR1*	44	TYR-41
AF-A0A8S8Y833-F1	Ribose 5-phosphate isomerase A *Candidatus Poseidoniales archaeon*	47	TYR-17, TYR-34
AF-A0A8S8YJI6-F1	N-acetyltransferase *Candidatus Poseidoniales archaeon*	36	TYR-27
AF-A0A8S8YVN2-F1	tRNA-binding domain-containing protein *Candidatus Poseidoniales archaeon*	48	PHE-16, PHE-20
AF-A0A8T4DH60-F1	NFYB/HAP3 family transcription factor subunit *Methanomicrobia archaeon*	46	TYR-16, HIS-30
AF-A0A8T5JI94-F1	Uncharacterized protein *Candidatus Bathyarchaeota archaeon*	45	PHE-28
AF-A0A8T5KDM7-F1	Desulfoferrodoxin FeS4 iron-binding domain-containing protein *Candidatus Bathyarchaeota archaeon*	38	TYR-8
AF-A0A8T6SHV5-F1	Class I SAM-dependent methyltransferase *Thermoproteota archaeon*	33	TYR-20
AF-A0A8T6TUW3-F1	RNA polymerase subunit sigma-24 *Nitrosopumilaceae archaeon*	33	PHE-19
AF-A0A8T6U953-F1	Ribose-phosphate pyrophosphokinase *Nitrosopumilaceae archaeon*	49	HIS-6
AF-A0A8T6UP66-F1	Thiolase N-terminal domain-containing protein *Candidatus Bathyarchaeota archaeon*	50	TRP-4, PHE-31
AF-A0A8T6WDJ4-F1	Methyltransferase domain-containing protein *Thorarchaeota archaeon (strain OWC)*	33	TYR-20
AF-A0A8T6XLB1-F1	tRNA-binding protein *Nitrosopumilaceae archaeon*	44	PHE-29, TRP-30
AF-A0A949JHM8-F1	DUF424 family protein *Theionarchaea archaeon*	50	HIS-44
AF-A1TJT8-F1	Large ribosomal subunit protein bL36 *Paracidovorax citrulli (strain AAC00-1)*	37	HIS-32

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-A1TYL8-F1	Large ribosomal subunit protein bL36 *Marinobacter nauticus (strain ATCC 700491 / DSM 11845 / VT8)*	37	HIS-32
AF-A1V881-F1	Large ribosomal subunit protein bL36 *Burkholderia mallei (strain SAVP1)*	38	HIS-33
AF-A1VES0-F1	Large ribosomal subunit protein bL34 *Nitratidesulfovibrio vulgaris (strain DP4)*	44	TYR-5, PHE-18
AF-A1WVA1-F1	Large ribosomal subunit protein bL36 *Halorhodospira halophila (strain DSM 244 / SL1)*	37	HIS-32
AF-A2S7J8-F1	Large ribosomal subunit protein bL36 *Burkholderia mallei (strain NCTC 10229)*	38	HIS-33
AF-A3MRX6-F1	Large ribosomal subunit protein bL36 *Burkholderia mallei (strain NCTC 10247)*	38	HIS-33
AF-A3P091-F1	Large ribosomal subunit protein bL36 *Burkholderia pseudomallei (strain 1106a)*	38	HIS-33
AF-A4VHQ1-F1	Large ribosomal subunit protein bL36 *Stutzerimonas stutzeri (strain A1501)*	38	HIS-33
AF-A4WFA7-F1	Large ribosomal subunit protein bL36A *Enterobacter sp. (strain 638)*	38	HIS-33
AF-A5CCQ4-F1	Large ribosomal subunit protein bL36 *Orientia tsutsugamushi (strain Boryong)*	41	PHE-27, PHE-36
AF-A5FMZ9-F1	Large ribosomal subunit protein bL36 *Flavobacterium johnsoniae (strain ATCC 17061 / DSM 2064 / JCM 8514 / BCRC 14874 / CCUG 350202 / NBRC 14942 / NCIMB 11054 / UW101)*	38	TYR-24, PHE-33
AF-A5FRW8-F1	Large ribosomal subunit protein bL36 *Dehalococcoides mccartyi (strain ATCC BAA-2100 / JCM 16839 / KCTC 5957 / BAV1)*	37	HIS-32
AF-A5GN09-F1	Large ribosomal subunit protein bL34 *Synechococcus sp. (strain WH7803)*	45	PHE-19, HIS-26
AF-A5GRN4-F1	Large ribosomal subunit protein bL34 *Synechococcus sp. (strain RCC307)*	45	PHE-19, HIS-26
AF-A6QCS0-F1	Large ribosomal subunit protein bL36 *Sulfurovum sp. (strain NBC37-1)*	37	HIS-32
AF-A6W371-F1	Large ribosomal subunit protein bL36 *Marinomonas sp. (strain MWYL1)*	37	HIS-32
AF-A7MPF5-F1	Large ribosomal subunit protein bL36A *Cronobacter sakazakii (strain ATCC BAA-894)*	38	HIS-33
AF-A7MWH9-F1	Large ribosomal subunit protein bL36A *Vibrio campbellii (strain ATCC BAA-1116)*	37	HIS-32
AF-A7ZSI8-F1	Large ribosomal subunit protein bL36A *Escherichia coli O139:H28 (strain E24377A / ETEC)*	38	HIS-33
AF-A8A5A4-F1	Large ribosomal subunit protein bL36A *Escherichia coli O9:H4 (strain HS)*	38	HIS-33
AF-A8ETK2-F1	Large ribosomal subunit protein bL36 *Aliarcobacter butzleri (strain RM4018)*	37	HIS-32
AF-A9ADL5-F1	Large ribosomal subunit protein bL36 *Burkholderia multivorans (strain ATCC 17616 / 249)*	38	HIS-33
AF-B1GZA6-F1	Large ribosomal subunit protein bL36 *Endomicrobium trichonymphae*	37	HIS-32
AF-B1IQ00-F1	Large ribosomal subunit protein bL36A *Escherichia coli (strain ATCC 8739 / DSM 1576 / NBRC 3972 / NCIMB 8545 / WDCM 00012 / Crooks)*	38	HIS-33
AF-B1JIY2-F1	Large ribosomal subunit protein bL36A *Yersinia pseudotuberculosis serotype O:3 (strain YPIII)*	38	HIS-33
AF-B1JU44-F1	Large ribosomal subunit protein bL36 *Burkholderia orbicola (strain MC0-3)*	38	HIS-33
AF-B1LHB3-F1	Large ribosomal subunit protein bL36A *Escherichia coli (strain SMS-3-5 / SECEC)*	38	HIS-33

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-B1WWJ3-F1	Large ribosomal subunit protein bL34 *Crocosphaera subtropica (strain ATCC 51142 / BH68)*	45	PHE-19, HIS-41
AF-B1X6F1-F1	Large ribosomal subunit protein bL36A *Escherichia coli (strain K12 / DH10B)*	38	HIS-33
AF-B3CQF1-F1	Large ribosomal subunit protein bL36 *Orientia tsutsugamushi (strain Ikeda)*	41	PHE-27, PHE-36
AF-B5FGD7-F1	Large ribosomal subunit protein bL36 *Aliivibrio fischeri (strain MJ11)*	37	HIS-32
AF-B8DP14-F1	Large ribosomal subunit protein bL34 *Nitratidesulfovibrio vulgaris (strain DSM 19637 / Miyazaki F)*	45	TYR-6, PHE-19
AF-B8GV37-F1	Large ribosomal subunit protein bL36 *Thioalkalivibrio sulfidiphilus (strain HL-EbGR7)*	37	HIS-32
AF-B8HR51-F1	Large ribosomal subunit protein bL34 *Cyanothece sp. (strain PCC 7425 / ATCC 29141)*	45	PHE-19
AF-C0HLA1-F1	Thrombin-like enzyme LmrSP-2 *Lachesis muta rhombeata*	30	HIS-12, PHE-14, TYR-19, PHE-24, PHE-25
AF-C6C1A7-F1	Large ribosomal subunit protein bL36 *Maridesulfovibrio salexigens (strain ATCC 14822 / DSM 2638 / NCIMB 8403 / VKM B-1763)*	37	HIS-32
AF-H0AEA3-F1	Metalloenzyme domain-containing protein *Haloredivivus sp. (strain G17)*	34	PHE-22
AF-I7CSI0-F1	Large ribosomal subunit protein eL39 *Natrinema sp. (strain J7-2)*	50	TRP-25, TRP-43
AF-M0B978-F1	Large ribosomal subunit protein eL39 *Natrialba aegyptia DSM 13077*	50	TRP-25, TRP-43
AF-M0BL78-F1	Short-chain dehydrogenase/reductase SDR *Halovivax asiaticus JCM 14624*	35	PHE-26
AF-M1P6N2-F1	Desulforedoxin *Methanosarcina mazei Tuc01*	41	TYR-10
AF-O96797-F1	Photosystem II reaction center protein Psb30 *Skeletonema costatum*	34	TRP-4
AF-P00273-F1	Desulforedoxin *Megalodesulfovibrio gigas*	37	TYR-8
AF-P0A404-F1	Photosystem I reaction center subunit XII *Synechococcus elongatus*	31	TYR-9, PHE-23, TYR-30
AF-P0A497-F1	Large ribosomal subunit protein bL36A *Vibrio cholerae serotype O1 (strain ATCC 39315 / El Tor Inaba N16961)*	37	HIS-32
AF-P0A7R0-F1	Large ribosomal subunit protein bL36 *Shigella flexneri*	38	HIS-33
AF-P17684-F1	Conantokin-T *Conus tulipa*	21	TYR-5
AF-P80502-F1	Chaperonin HSP60, mitochondrial *Solanum tuberosum*	40	PHE-7
AF-P83867-F1	Ocellatin-3 *Leptodactylus ocellatus*	21	HIS-16
AF-P84337-F1	Homotarsinin *Phyllomedusa tarsius*	24	HIS-12
AF-P84477-F1	Myofibril-bound serine protease *Cyprinus carpio*	22	TYR-5, HIS-10, TRP-14, PHE-17
AF-P84807-F1	Cysteine-rich venom protein 25-A *Naja haje haje*	29	PHE-4, HIS-22
AF-P86498-F1	Keratin *Cervus elaphus*	23	TYR-1
AF-Q0ABF4-F1	Large ribosomal subunit protein bL36 *Alkalilimnicola ehrlichii (strain ATCC BAA-1101 / DSM 17681 / MLHE-1)*	37	HIS-32
AF-Q0T003-F1	Large ribosomal subunit protein bL36 *Shigella flexneri serotype 5b (strain 8401)*	38	HIS-33

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
AF-Q0TCG2-F1	Large ribosomal subunit protein bL36A *Escherichia coli O6:K15:H31 (strain 536 / UPEC)*	38	HIS-33
AF-Q0VSI2-F1	Large ribosomal subunit protein bL36 *Alcanivorax borkumensis (strain ATCC 700651 / DSM 11573 / NCIMB 13689 / SK2)*	38	HIS-33
AF-Q1BRX0-F1	Large ribosomal subunit protein bL36 *Burkholderia orbicola (strain AU 1054)*	38	HIS-33
AF-Q1R632-F1	Large ribosomal subunit protein bL36A *Escherichia coli (strain UTI89 / UPEC)*	38	HIS-33
AF-Q1XDA2-F1	Photosystem II reaction center protein Psb30 *Pyropia yezoensis*	34	TRP-4
AF-Q21M37-F1	Large ribosomal subunit protein bL36 *Saccharophagus degradans (strain 2-40 / ATCC 43961 / DSM 17024)*	38	HIS-33
AF-Q21QP5-F1	Large ribosomal subunit protein bL36 *Albidiferax ferrireducens (strain ATCC BAA-621 / DSM 15236 / T118)*	37	HIS-32
AF-Q2JHS2-F1	Large ribosomal subunit protein bL34 *Synechococcus sp. (strain JA-2-3B'a(2-13))*	46	PHE-19, TYR-41
AF-Q2JI37-F1	Photosystem I reaction center subunit XII *Synechococcus sp. (strain JA-2-3B'a(2-13))*	30	PHE-7
AF-Q2JT51-F1	Photosystem I reaction center subunit XII *Synechococcus sp. (strain JA-3-3Ab)*	30	PHE-7
AF-Q30TS7-F1	Large ribosomal subunit protein bL36 *Sulfurimonas denitrificans (strain ATCC 33889 / DSM 1251)*	37	HIS-32
AF-Q318K4-F1	Large ribosomal subunit protein bL36 *Prochlorococcus marinus (strain MIT 9312)*	38	TYR-24
AF-Q31VX7-F1	Large ribosomal subunit protein bL36 *Shigella boydii serotype 4 (strain Sb227)*	38	HIS-33
AF-Q32B52-F1	Large ribosomal subunit protein bL36 *Shigella dysenteriae serotype 1 (strain Sd197)*	38	HIS-33
AF-Q3AM28-F1	Large ribosomal subunit protein bL34 *Synechococcus sp. (strain CC9605)*	45	PHE-19, HIS-26
AF-Q3AV42-F1	Large ribosomal subunit protein bL34 *Synechococcus sp. (strain CC9902)*	45	PHE-19, HIS-26
AF-Q3YWW0-F1	Large ribosomal subunit protein bL36 *Shigella sonnei (strain Ss046)*	38	HIS-33
AF-Q3Z957-F1	Large ribosomal subunit protein bL36 *Dehalococcoides mccartyi (strain ATCC BAA-2266 / KCTC 15142 / 195)*	37	HIS-32
AF-Q3ZZQ0-F1	Large ribosomal subunit protein bL36 *Dehalococcoides mccartyi (strain CBDB1)*	37	HIS-32
AF-Q46501-F1	Large ribosomal subunit protein bL36 *Oleidesulfovibrio alaskensis (strain ATCC BAA-1058 / DSM 17464 / G20)*	37	HIS-32
AF-Q47J81-F1	Large ribosomal subunit protein bL36 *Dechloromonas aromatica (strain RCB)*	37	HIS-32
AF-Q4A8J4-F1	Large ribosomal subunit protein bL36 *Mesomycoplasma hyopneumoniae (strain 7448)*	37	HIS-32
AF-Q4AAG3-F1	Large ribosomal subunit protein bL36 *Mesomycoplasma hyopneumoniae (strain J / ATCC 25934 / NCTC 10110)*	37	HIS-32
AF-Q4G392-F1	Large ribosomal subunit protein bL34c *Emiliania huxleyi*	45	HIS-7, PHE-19
AF-Q5E893-F1	Large ribosomal subunit protein bL36 *Aliivibrio fischeri (strain ATCC 700601 / ES114)*	37	HIS-32

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AlphaFold DB model	Protein / source organism	Residues	Residues replaced by alanine
`AF-Q5L8D2-F1`	Large ribosomal subunit protein bL36 *Bacteroides fragilis (strain ATCC 25285 / DSM 2151 / CCUG 4856 / JCM 11019 / LMG 10263 / NCTC 9343 / Onslow / VPI 2553 / EN-2)*	38	TYR-24, TYR-33
`AF-Q5N606-F1`	Large ribosomal subunit protein bL34 *Synechococcus sp. (strain ATCC 27144 / PCC 6301 / SAUG 1402/1)*	45	PHE-19
`AF-Q62GM7-F1`	Large ribosomal subunit protein bL36 *Burkholderia mallei (strain ATCC 23344)*	38	HIS-33
`AF-Q661C1-F1`	Large ribosomal subunit protein bL36 *Borrelia garinii subsp. bavariensis (strain ATCC BAA-2496 / DSM 23469 / PBi)*	37	HIS-32
`AF-Q664U2-F1`	Large ribosomal subunit protein bL36A *Yersinia pseudotuberculosis serotype I (strain IP32953)*	38	HIS-33
`AF-Q6B8L0-F1`	Photosystem II reaction center protein Psb30 *Gracilaria tenuistipitata var. liui*	34	TRP-4
`AF-Q6KI32-F1`	Large ribosomal subunit protein bL36 *Mycoplasma mobile (strain ATCC 43663 / 163K / NCTC 11711)*	38	HIS-33
`AF-Q6YQZ8-F1`	Large ribosomal subunit protein bL36 *Onion yellows phytoplasma (strain OY-M)*	38	TYR-24, HIS-33
`AF-Q72D55-F1`	Large ribosomal subunit protein bL34 *Nitratidesulfovibrio vulgaris (strain ATCC 29579 / DSM 644 / CCUG 34227 / NCIMB 8303 / VKM B-1760 / Hildenborough)*	44	TYR-5, PHE-18
`AF-Q73PL1-F1`	Large ribosomal subunit protein bL36 *Treponema denticola (strain ATCC 35405 / DSM 14222 / CIP 103919 / JCM 8153 / KCTC 15104)*	37	HIS-32
`AF-Q7MTN6-F1`	Large ribosomal subunit protein bL36 *Porphyromonas gingivalis (strain ATCC BAA-308 / W83)*	38	TYR-24, TYR-33
`AF-Q8ZJ91-F1`	Large ribosomal subunit protein bL36A *Yersinia pestis*	38	HIS-33
`AF-Q98Q05-F1`	Large ribosomal subunit protein bL36 *Mycoplasmopsis pulmonis (strain UAB CTIP)*	37	HIS-32
`AF-Q9TLX5-F1`	Photosystem I reaction center subunit XII *Cyanidium caldarium*	30	PHE-8, PHE-22, TYR-29
`AF-Q9UR66-F1`	Nucleoside diphosphate kinase *Candida albicans*	21	HIS-4, TRP-19, PHE-20
`AF-U1PBV3-F1`	Acyl-CoA transferases/carnitine dehydratase *halophilic archaeon J07HX5*	47	TYR-19